



Wheel–rail-induced derailment analysis: a comprehensive literature review of experimental and simulation-based approaches

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Abstract

This paper offers a comprehensive review of the literature on the role of wheel–rail contact mechanics in train derailments. Using experimental and simulation methods, this review examines various research efforts that analyze how contact forces influence wheel climb and derailment dynamics. The related studies are summarized, and insights are provided on how they have contributed to understanding derailments and enhancing overall rolling stock safety. The review shows significant progress across different specialized areas within the broader topic of derailments. This includes advanced, state-of-the-art testing rigs, high-fidelity models that accurately replicate field conditions, materials that help prevent derailments, and wheel and rail profiles that reduce derailment risks. However, the accuracy of testing and modeling often requires more complex setups, sophisticated data analysis techniques, and greater resources. Despite these advances over the past few decades, further scientific research is necessary to understand better the root causes of events like wheel climb derailments under controlled and repeatable conditions.

Keywords Wheel–rail interaction · Derailments · Roller rig · Safety · Railroad industry · Simulation · FEM · Analytical · Nadal’s derailment coefficient · Wheel climb · Contact forces · Wheel and rail defects

1 Introduction

The constantly evolving railway industry has played a crucial role in shaping global transportation, economies, and societies. A nation’s prosperity is closely linked to the growth and development of its rail sector. The railway concept can be traced back to early civilizations, dating back to around the sixth century, when makeshift tracks were created in grooves to carry heavy loads [1]. As modernization

progressed, between 1801 and 1804, British engineer Richard Trevithick built the world’s first steam-powered locomotive [2]. This remarkable achievement propelled transportation technology far ahead of its time. Americans recognized the potential benefits of rail-based transportation when they evaluated the steam locomotive as a power source. This led to the founding of the first railroad around 1827 [3]. Today, the rail networks and track lengths of these countries extend extensively across their entire landscapes. The USA has approximately 160,000 miles of track, with freight transportation totaling approximately 2863 billion tonne-kilometers. Likewise, Europe and Asia have track lengths of 126,000 miles and 100,000 miles, with freight capacities of 400 billion tonne-kilometers and 4.5 trillion tonne-kilometers, respectively [4–6].

The railroad industry has been used for transporting passengers (intercity, regional, and commuter) and freight (intermodal, flatcars, and boxcars). In the USA, the majority has been the latter, whereas in the rest of the world, there is a healthy mix of freight and passenger transportation, varying in percentages by country and geography [7]. Railroads are among the most efficient forms of transportation, second only to maritime transport. A train can carry one ton

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Fig. 1 BNSF railway derailment (courtesy of BNSF) [11]

of freight for about 500 miles on just one gallon of fuel, offering superior “ton-mile” efficiency compared to other modes of transport [8, 9].

Similarly, railroads are considered one of the safest modes of transportation in terms of economic losses and other unfortunate incidents, such as human fatalities. In countries such as the USA, Great Britain, and Europe, significant improvements have been made to enhance the industry’s safety over the last few decades. It has been observed that most fatal collisions and derailments occur at either grade crossings (also known as level crossings) or due to illegal trespassing [10]. Another factor affecting railroad safety is the operating environment. For instance, top-of-rail contaminations due to fallen leaves, oil, or grease can negatively affect traction, resulting in longer stopping distances during braking or difficulty generating traction on a steep grade. The BNSF railway derailment (Fig. 1) resulted in one fatality and damages worth \$15.6 million [11]. Similarly, a trolley derailment that occurred in Southeastern Pennsylvania Transportation in July 2023 incurred damages worth \$800,000 [12]. Lastly, the CSX Transportation head-on collision in April 2024 resulted in three injuries and damages worth \$720,000 [13]. Based on these examples, it is clear that the dire costs and injuries caused by such unforeseen incidents necessitate immediate measures. Major countries and the private sector must adopt a zero-accident policy to take appropriate actions that consider safety requirements and promote further industry development [14].

Due to the freight sector’s high potential to positively impact the economy, the government has traditionally regulated and overseen rulemaking for the railroad industry. In the USA, the Federal Railroad Administration (FRA) establishes and enforces standards that must be adhered

to nationwide. For example, an FRA mandate, such as the positive train control (PTC) [15], helps prevent collisions and derailments, thus improving rail safety. Similar agencies exist, such as the European Union Agency for Railways (ERA), the Indian Railway’s RDSO, and the Association of the Southeast Asian Nations (ASEAN). These government agencies take proactive measures to significantly enhance rail safety. To foster more competitive dynamics, improve efficiency, and align with the evolving trends in technology, the privatization of rail networks is also crucial. With the Staggers Rail Act of 1980, [16] privatization and deregulation emerged in the USA. The principal railways have been with the private sector, including the Class 1 railroads such as Norfolk Southern (NS), BNSF Railway, CSX Transportation, and Union Pacific. Similar privatization has been observed in the United Kingdom and Japan.

To improve the holistic efficiency of the railroad and, subsequently, the safety of operation, one must address several operational challenges that the industry faces while a train is in motion. These challenges include the possibility of collisions due to increased train operations on congested networks, infrastructure degradation, and derailments that occur for several reasons, such as wheel–rail interaction, wheelset or axle defects, high speeds, switch and crossing faults, and low adhesion, which increases braking distance [17–19]. These are some of the causes of high interest and can be considered a trending topic that several highly motivated researchers are currently focusing on. Although there has been substantial research across all the above-mentioned challenges, derailments resulting from wheel–rail interaction—flange climbing, gauge widening, and wheel or rail defects—are prevalent globally. As highlighted earlier, these lead to financial losses, environmental damage, and a

deterioration in the public perception of rail safety. Due to the complexity of the wheel–rail contact interface, a global research effort is imperative. In this regard, significant research has been conducted experimentally in a laboratory environment and through field tests, using several analytical and numerical models developed over the past few decades.

This paper reviews the significant advances made in the past 2 decades in the context of wheel–rail interaction-induced derailments. It examines how they have contributed to increasing the current knowledge and, thus, the safety of the railroad industry. It analyzes the importance of experimental testing apparatus and simulations, as they complement each other when comparing results, thereby validating both. An overview of different testing methodologies is also provided to support the results. This article aims to provide a comprehensive understanding through empirical insights, methods adopted, and gaps in wheel–rail interaction-based derailment research, to inspire future technologies that improve safety and advance railway engineering sciences. To facilitate a thorough analysis of several published manuscripts, a detailed methodology is adopted to screen for papers well suited for this review article. To provide a brief pictorial representation of the method adopted in structuring

this article, a flowchart is added to summarize it, as shown in Fig. 2. It is essential to note that this review effort focuses on the progress in experimental and simulation-based understanding of WRI and derailments, rather than the broader scope of WRI and derailments, as this topic encompasses a much wider range of knowledge and expertise.

2 Derailments: types and occurrence

2.1 Derailment and its types

A train is derailed when one or more rail vehicle wheels leave the track, i.e., the wheel–rail contact is broken. This disrupts regular operations, leading to significant damage and injuries. This statement might portray derailment as a well-understood problem, but does not mislead by it. Train derailments have been a problem for centuries. Researchers and operators worldwide have expressed concern about train derailments. One must always go to the root cause to understand the physics behind it. Jun and Qingyuan [20] investigated the mechanical mechanism of train derailments. The system's transverse vibration is deemed stable when

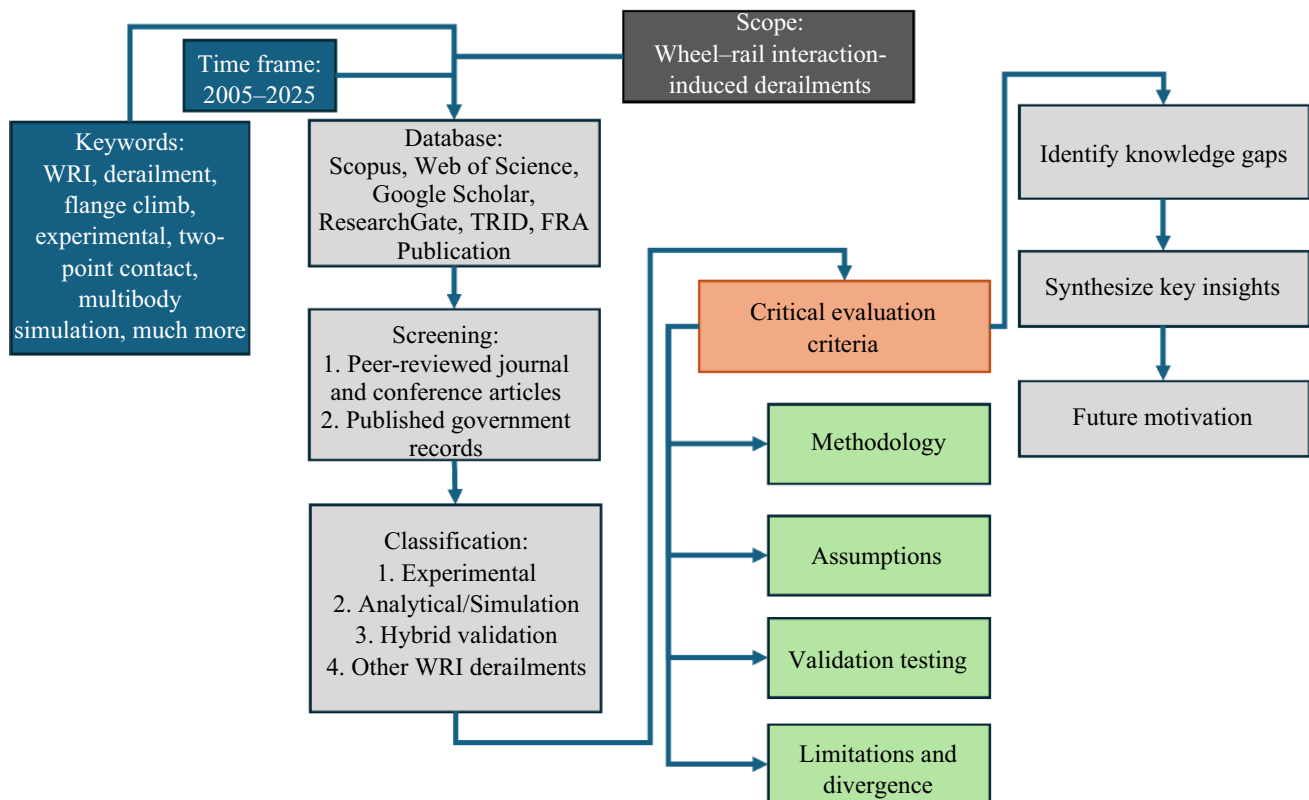


Fig. 2 Methodology adopted for this literature review

it exhibits snake-like behavior, characterized by continuous oscillating contact between the flange and the rail, such as hunting-type oscillations. Now, in an unstable situation where only the flange of one wheel rises to the top, the transverse movement is restricted, thereby delaying the transverse vibration as the wheel loses stability. The authors provide an in-depth assessment of the random energy associated with train derailment. They consider some real case scenarios to prove that the derailment coefficient and transverse loads are incredibly high and that derailments are instantaneous (happen within 0.2 s). The authors emphasize that theoretical derailment analysis is a complex process that requires additional effort. In the past, only an initial survey was conducted by Blader in 1990 [21]. Much more sophisticated efforts are necessary. Now, every mechanical system is governed by the fundamentals; the dynamics of train derailment can also be explained using a set of equations of motion. One such example of the dynamic equations is provided by Jeong et al. [22], who compare their results from FORTRAN to those of the ADAMS (commercial) model. The comparison showed reasonable verification, which future researchers can use to tune the model with other parameters and improve efficiency.

Derailment can be classified into different types based on causes, mechanisms, and operational conditions. FRA identifies these causes and categorizes them as train defects, wheel defects, rolling stock failures, human errors, signal faults, and many others [23, 24]. Understanding these is essential for developing mitigation strategies. Below are a

few types of derailments (Fig. 3), listed by their causes to provide a general context and some examples:

- *Wheel climb derailment* It occurs when a wheel flange climbs over the railhead instead of following the intended path.
- *High/low-speed derailment* These derailments occur when trains operate at excessive speeds, particularly around curves, causing instability and track overturning. Low speeds occur in yards, stations, or low-speed turnouts. These derailments are often attributed to track defects, improper switch settings, or loss of wheel–rail adhesion.
- *Track irregularity-induced derailment* Track defects such as misalignment, broken rails, gauge widening, or settlement can cause the wheelset to lose proper guidance, resulting in derailment.
- *Collision-induced derailment* If a train collides with an obstacle, such as a vehicle at a level crossing or another train, the impact force can dislodge the wheels from the rails.
- *Human factor-related derailment* Signal misinterpretations or improper train handling can lead to derailment scenarios.

As this list is not exhaustive, this article primarily focuses on derailments induced by wheel–rail interaction, analyzing the influence of wheel or flange climb, lateral/vertical force ratios (derailment coefficient), gauge widening, tractive



Fig. 3 Different types of derailment recorded

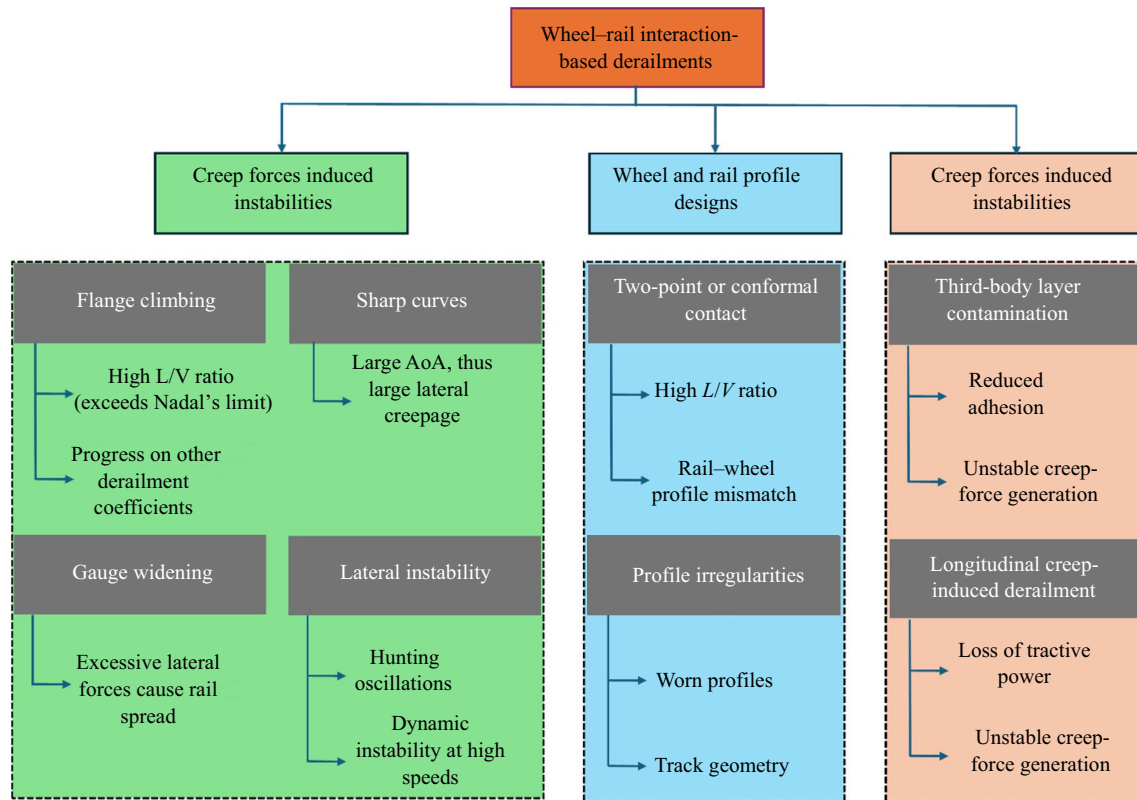


Fig. 4 Wheel–rail interaction (WRI)-induced derailment types

(adhesive) forces (contact forces), and track buckling. Figure 4 illustrates a flowchart that outlines various causes of derailments resulting from wheel–rail interaction. Although this is not an exhaustive list, the research on understanding the causes of derailments is limited by the difficulties researchers face in comprehending the complexity of WRI. This creates a gap in thoroughly understanding derailments, thus requiring a significant amount of work to be done. Based on these and several others recorded, some key factors that play an influential role in derailment can also be categorized such as wheel–rail interaction, lateral instability (hunting), speed or acceleration, curving factors, human factors, and environmental factors [17, 25–28]. Wilson et al. [29] analyzed and presented a more in-depth comparative review of various derailment mechanisms and current state-of-the-art assessment methods. This study critically examines experimental and simulation-based approaches to derailment analysis, exploring their role in enhancing railroad safety and operational efficiency. This work highlights how ongoing research drives improvements in industry standards and practices by discussing key methodologies and their contributions. Furthermore, this study identifies key research gaps and discusses potential strategies to bridge them, contributing to the advancement of derailment prevention and railway safety.

2.2 Statistical survey of derailment occurrence

Over the last few years, on average, there have been three reported incidents of daily derailment, totaling approximately one thousand derailments annually [30]. The national rail network covers almost 58% of all major cities. This means that, during unforeseen incidents, there are higher chances of cities and their populations being in the crossfire. Figure 5 shows the derailment hotspot commonly observed in the USA [31]

Cao et al. [32] conducted an extensive statistical analysis to study railway safety in China and Japan, using data from nearly 3 decades. In China, 29.67% of all accidents are due to derailments. The dataset contains approximately ninety-one cases, as there is no portal available that records and publishes all incidents/accidents that have occurred. Japan has experienced 346 train accidents over the last 2 decades, with 223 resulting from train derailments. This constitutes 64% of all safety incidents [33]. Similar numbers are estimated by Cao et al. [32], who report that the number of derailments observed over 3 decades accounts for 58.97% of all recorded incidents. Figure 6 summarizes the percentage of derailments that are recorded in these major railroad operating countries.

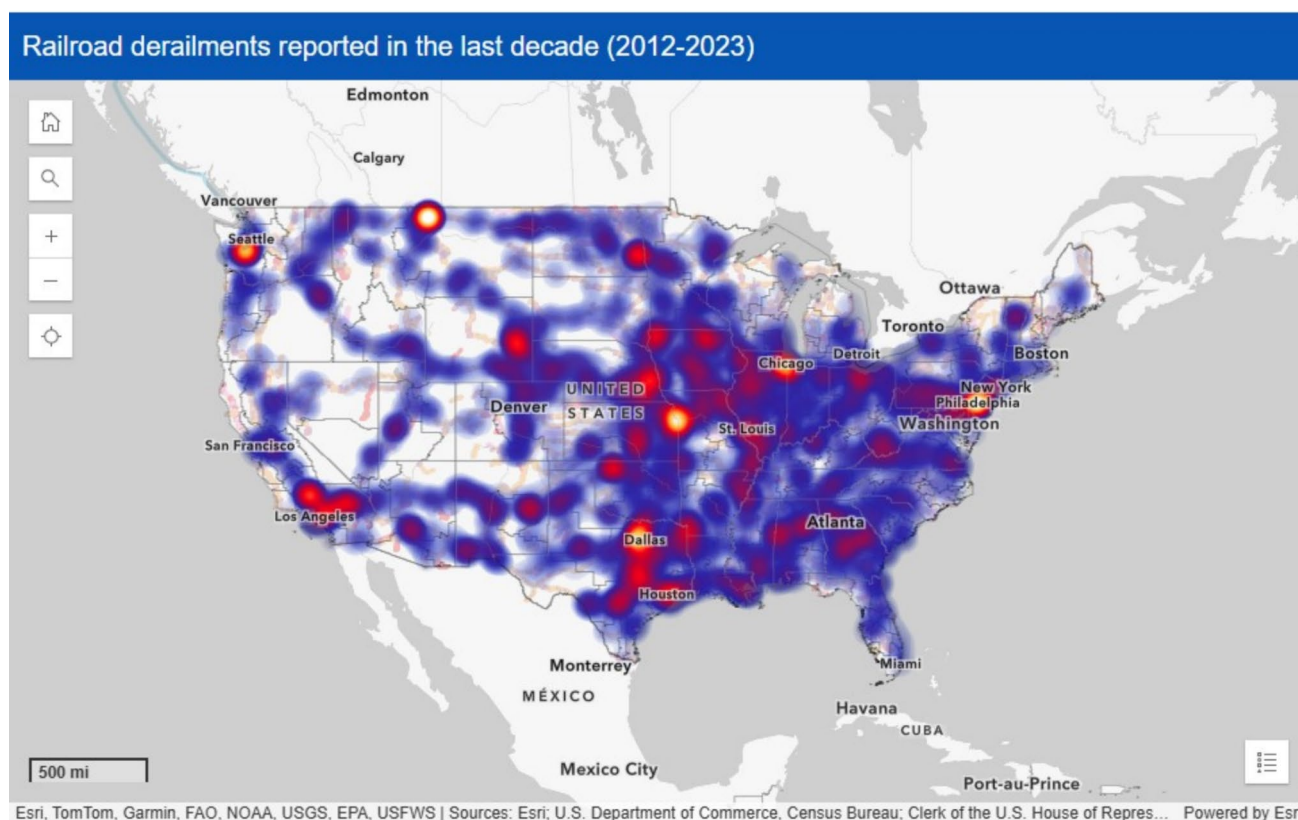


Fig. 5 Derailment hot zones in the USA [31]

3 Importance of wheel–rail contact

3.1 Hertzian contact theory

Heinrich Hertz developed a classical theory in 1882 [38], which has been widely used as a starting point for many contact models and theories. Hertzian contact theory calculates the contact area and pressure distribution following a normal (vertical) contact approach, provided that a set of restrictive assumptions is met. Some key assumptions related to the wheel–rail contact are:

1. Wheel and rail materials are homogeneous, and their properties are isotropic.
2. Wheel and rail contact is frictionless and perfectly elastic (no plastic deformation).
3. The kinematic equations and material elastic properties are linear.
4. The curvature of contact is constant within the contact patch.
5. The contact surface radii are much larger than the contact patch.

Considering these assumptions, the Hertz contact theory concludes that the contact area between the two

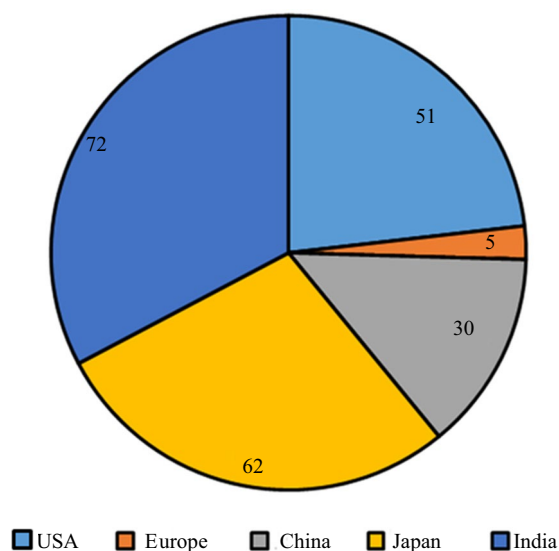


Fig. 6 Summary derailment incidents recorded percentage in major countries [34–37]

non-conformal bodies would be elliptical, as shown in Fig. 7.

This contact patch follows a semi-ellipsoid normal pressure distribution, defined in Eq. (1):

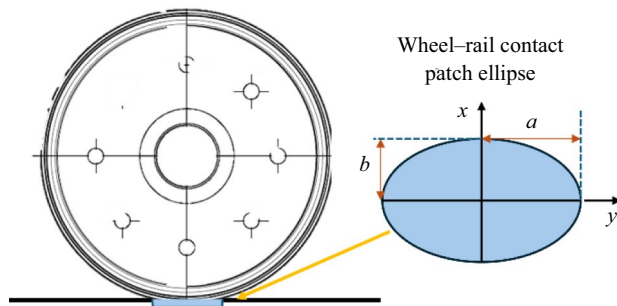


Fig. 7 Wheel–rail contact patch shape (enlarged for clarity) and dimensional nomenclature

$$p(x, y) = \frac{3N}{2\pi ab} \sqrt{1 - \left(\frac{x}{a}\right)^2 - \left(\frac{y}{b}\right)^2}, \quad (1)$$

where N is the normal load, a is the major axis, and b is the minor axis of the ellipse.

Hertzian contact theory is a good approximation for a single-point contact that can be measured for a cylindrical wheel profile or tapered tread with small contact angles. Radmehr et al. [39] take a similar approach to measuring the contact patch for the tread region of the AAR-1B wheel profile. Several non-Hertzian and semi-Hertzian normal contact theories can be more effectively applied to a wheel–rail normal contact problem. One such theory is Kalker's non-Hertzian model [40].

Kalker's exact three-dimensional (3D) rolling contact theory [41–43] is one of the most successful non-Hertzian models for applying wheel–rail geometry. The model presents the wheel and rail as elastic half-spaces with surface curvature radii much greater than the dimensions of the contact patch. The model can accurately predict the contact for cases of non-elliptical and multi-point contact patches. The numerical-based model is incorporated in a software called CONTACT [44]. The software can solve normal and tangential contact problems by interacting with several parameters as needed. More details on contact theories and contact models can be found in the comprehensive review articles by Meymand et al. [45] and Fang et al. [46].

The contact of a flange root and gauge corner results in conformal contact conditions. The flange and gauge corner, because of a high angle of attack (AoA), result in two-point contact conditions. These are some non-Hertzian cases that can be solved using the modified Kalker's theory to obtain conformal contact solutions, as is done in CONTACT [47, 48].

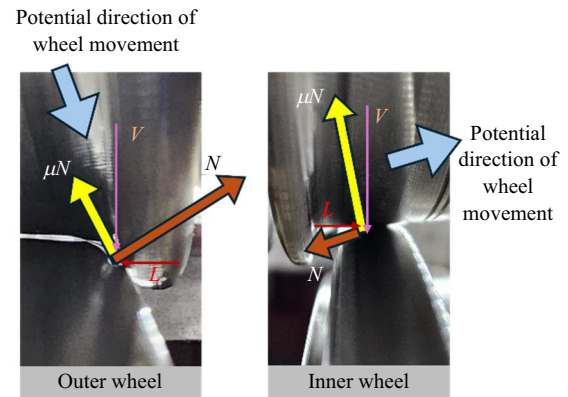


Fig. 8 Normal direction of movement of outer and inner wheel in a curve [72, 90]

3.2 Derailment criteria

Nadal provided a relationship between wheel flange forces in track coordinates and equivalent forces in contact coordinates [49]. The vertical forces are usually denoted as F_V and lateral forces as F_L , often simply abbreviated as V and L . These forces can be reconstructed into normal (N) and tangential (μN) components, as depicted in Fig. 8. The results of F_s and F_N are due to dynamic creep forces. The ratio between F_L and F_V (i.e., L/V) is called the Nadal's limit or the derailment coefficient and expressed in terms of wheel–rail parameters as:

$$\frac{F_L}{F_V} = \frac{\tan \Theta - \mu}{1 + \mu \tan \Theta}, \quad (2)$$

where Θ is the flange angle, which in certain cases is the contact angle, and μ is the coefficient of friction. Under saturated lateral creepage conditions, μ is numerically equal to the tangential traction coefficient (F_s). The Nadal criterion estimates that when the ratio in Eq. (2) is > 1 , derailment occurs, or the wheel is at the onset of wheel climb derailment.

Owing to several assumptions, such as not including the effects of yaw angle and contact angle, many researchers consider the Nadal criterion to be too conservative for estimating derailment, particularly at small angles of attack (AoA), as is verified by the test results shown in Fig. 9.

3.3 Unloading or off-loading ratio criteria

Wheel unloading (WUL) refers to the amount of static load (Q_0) removed from the wheel–rail interface (WRI) due to factors such as vehicle dynamics, track defects, and external

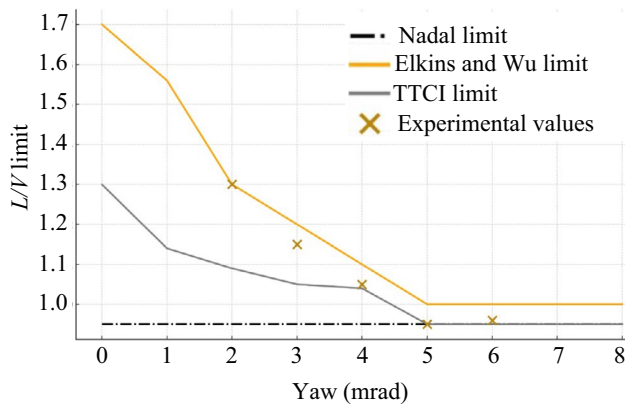


Fig. 9 Comparison of derailment limits showing the conservative nature of Nadal's derailment criteria at lower AoA's (adapted from [50, 55])

factors like wind. This is a commonly applied method to evaluate the wheel lift phenomenon in the Asian subcontinent. According to APTA standards, WUL can be estimated as a percentage loss of the nominal wheel load (NWL) on a displaced track [51]. Over the years, several researchers have evaluated various methods for predicting and measuring WUL ratios to correlate them with derailment mechanics and derailment coefficients.

The studies related to unloading or off-loading derailment predictions are summarized in Table 1. Zeng et al. [52] refined the indirect measurement approach for evaluating derailment safety by introducing a framework that linked the WUL ratio with the wheelset derailment coefficient (L/V) through a single “triangular safety zone.” This modification addressed a key limitation of earlier single-parameter derailment models, allowing both lateral force development and wheel load transfer to be considered together. The identified wheel–rail forces from the indirect method showed strong agreement with results from instrumented wheelsets (IWSs) and SIMPACK simulations, with peak deviations below 15%. Despite its effectiveness, the study recognized that assuming a rigid wheelset reduced accuracy at higher frequencies and that neglecting contact point shifts and elastic vibrations could affect results under complex loading. Even with these limitations, the study provided a significant step toward creating a more realistic and measurable derailment safety framework.

Liang et al. [53] developed a dynamic derailment simulation model to better understand how WUL contributes to train instability and potential derailment. Their study focused on capturing how changes in vertical wheel load, combined with lateral forces, influence wheel climb and loss of rail contact. Using a detailed multibody dynamic model, the authors simulated various track irregularities, speeds, and loading conditions. The results showed that the WUL

Table 1 Summarized reviews on unloading ratio criteria

Serial No.	Author (year)	Technique used	Scope	Remarks
1	Zeng et al. (2016) [52]	Simulation and experimental	Linked wheel unloading ratio and derailment coefficient under various dynamic conditions (crosswinds, hunting, light rail)	Balanced lateral and vertical load effects; limited by neglect of contact shifts; not exclusively for wheel–rail related derailment mechanisms
2	Liang et al. (2014) [54]	Simulation	Modeled vertical load loss and lateral forces to study transient derailment onset; focused on unloading ratio sensitivity	Rigid-body model; lacks experimental validation; simplifications limit field applicability
3	Pagaimo et al. (2020) [54]	Simulation and statistical modeling	Analyzed freight wagon derailment via joint L/V and $\Delta Q/Q_0$ indices under asymmetric loading and suspension faults	Validated numerically; limited by rigid-suspension assumption; requires more longitudinal dynamics
4	Azizi et al. (2020) [55]	3D finite element model	Quantified wheel unloading from track geometry defects and stiffness loss	Link between local track degradation and unloading; partial validation with field observations; limited by suspension coupling and damping
5	Gong et al. (2021) [56]	Full-scale experimental	Measured lateral/vertical forces and unloading ratio across speed and curvature conditions	High-quality validation confirms permissible limits and correlation between tight curves and high unloading; it lacks transient or nonlinear suspension assessment

ratio is a sensitive and reliable indicator of derailment potential, especially when analyzed alongside the traditional L/V ratio. However, the study was limited to numerical simulations without experimental validation, and certain simplifications restricted its applicability to field conditions. Despite these limitations, the work established an important connection between transient load reduction and derailment mechanisms.

Pagaimo et al. [54] presented a derailment investigation of freight wagons using a response surface methodology (RSM) to identify the conditions that maximize derailment risk. Their study combined multibody simulations in VAMPIRE with statistical optimization to assess the influence of asymmetric loading, speed, and suspension failures on WUL and derailment indices. Both WUL ratio ($\Delta Q/Q_0$) and L/V ratio were used jointly to evaluate safety margins, showing that derailment risk was highest when high-speed operation coincided with Lenoir link failure and cargo imbalance. The RSM predictions closely matched detailed simulations, validating the methodology's robustness. However, the approach remains limited by simplified suspension modeling and rigid-body assumptions. Still, this work advanced derailment prediction by combining mechanical modeling with statistical design, showing that unforeseen unloading conditions can precede derailment even when L/V ratios remain below established safety limits.

Azizi et al. [55] investigated wheel load reduction in railway tracks featuring unsupported sleepers and surface irregularities, highlighting how local stiffness loss can trigger WUL and increase derailment potential. Using a 3D finite element (FE) model, they evaluated transient wheel–rail interactions to quantify the vertical load transfer and its relationship to track geometry defects. The study effectively demonstrated that WUL increases sharply with larger unsupported spans and deeper surface irregularities, often preceding unsafe L/V ratios. Although primarily simulation based, their results aligned well with field-measured trends from previous literature, providing partial validation. The main limitation lies in the simplified boundary conditions that exclude vehicle suspension dynamics and real damping effects, which may underestimate the combined influence of track–vehicle coupling. It is worth noting that their findings established a strong link between local track degradation, an increase in the unloading ratio, and an increased likelihood of derailment.

Gong et al. [56] conducted a full-scale experimental study on a metro bogie operating at high speeds to analyze wheel–rail interaction forces and assess derailment safety in line with UIC 518 standards. Using IWSs, they measured lateral and vertical forces to calculate the derailment coefficient and WUL ratio under both no-load and overload conditions. The results, validated through direct measurements, showed that both indices peaked on the tighter

curves, suggesting exacerbated unloading and lateral load transfer. The analysis, however, was limited to steady-state conditions without dynamic modeling or evaluation of transient excitation. While the study provided valuable empirical data linking unloading ratio and derailment coefficient, it did not explore nonlinear suspension effects or high-frequency dynamics that often contribute to real derailment events.

It is worth noting that several of these studies highlight the importance of using newer criteria in correlation to Nadal's established derailment criteria. They suggest the need for more understanding of such approaches using other—potentially more capable—validation tools, both experimental and numerical.

4 Variations to Nadal's derailment criterion

Since Nadal's proposed criterion, a handful of researchers have attempted to revise it, mostly dating between 1963 and 1999 [57, 58]. These include the studies by Gilchrist et al. [59] in 1976, and Sweet et al. [60] in early 1980s. In the last couple of decades, additional researchers have taken inspiration, thereby providing further interpretations, calculations, and dynamic models to advance the criteria by adding additional parameters or making it less conservative.

Barbosa et al. [61] use a moving tri-orthogonal reference on a wheelset. A set of basic dynamic equations using Newton's laws is transformed by rotating the matrix about the contact angle and the AoA. Based on this, three quasi-static equilibrium equations are derived for lateral, vertical, and moment. Thus, the L/V ratio becomes a function of the contact angle, AoA (σ), and two tangential force components, as shown below:

$$\frac{L}{V} = \frac{A\mu - B \tan \alpha}{B + A\mu \tan \alpha}. \quad (3)$$

The author formulates a new flange climb criterion by following Coulomb's friction laws and the tangential contact problem from Kalker's linear theories in [42]. The coefficients added to Nadal's equation are

$$A = k_x \sin \psi + (1 + k_y) \cos \psi, \quad (4)$$

$$B = \sqrt{k_x^2 + 1 + 2k_y + k_y^2}. \quad (5)$$

As the above equations show, the term A depends on the yaw angle, and B is purely force dependent. Additionally, k_x is the ratio between longitudinal/transverse forces, and k_y is the ratio between the couple/transverse forces.

This new expression, which is compatible with Nadal's conservative coefficient, yields a three-dimensional rolling contact parameterization for different vehicle dynamic

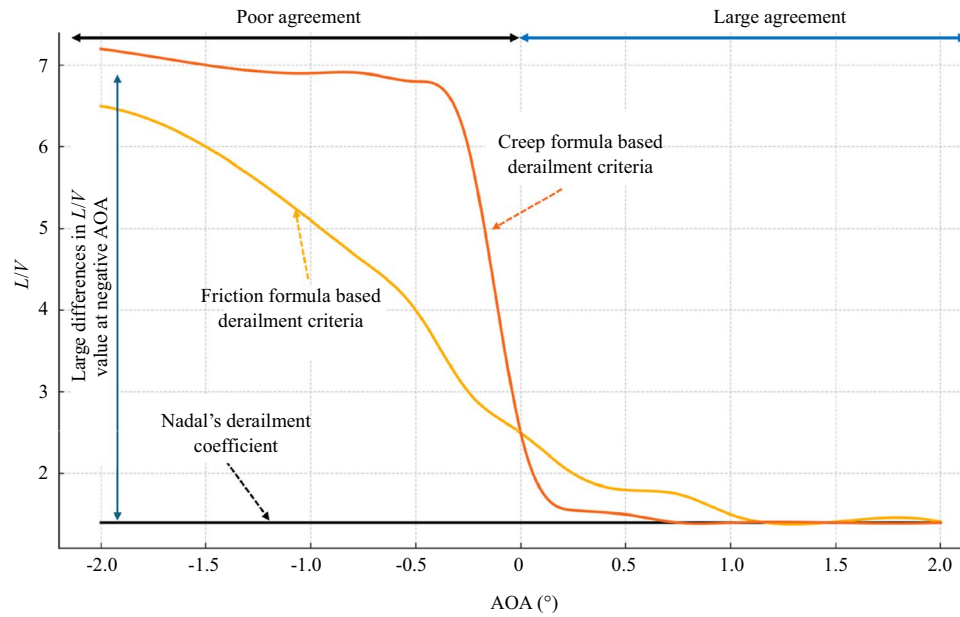


Fig. 10 Comparison of different derailment criteria showing large deviations from Nadal's criteria at negative AoA, while matching fairly well at positive AoA (adapted from [63])

scenarios. Shu et al. [62] have developed a bi-parameter flange climb distance criterion for the AAR-1B and AREMA-136-pound rail profiles, which is restricted to only these profiles. The authors use the influence of AoA and the climb distance (D) in their nonlinear equations although they conducted validation tests for specific applications. Notably, the tests were for $\text{AoA} > 5$ mrad, which introduce some degree of conservatism in the derived expressions. Similarly, Zeng et al. [63] simplified the formulation with the effects of AoA, contact angle, and the coefficient of friction. The authors propose two derailment criteria based on a quasi-static model of a wheelset, Coulomb's friction laws, and Kalker's linear theory for creep force assumptions. They simplified the criterion in Eqs. (6)–(9) to a single equation, namely

$$\frac{L}{V} = c(\psi) \frac{\tan \delta_L - \mu}{1 + \mu \tan \delta_L}. \quad (6)$$

The $c(\psi)$ term is a function of AoA and the friction coefficients:

$$c(\psi) = c_1 - c_2 \arctan(c_3 \psi + c_4). \quad (7)$$

More practical tests in an experimental setup are required to further validate the use of this simplified equation. The results show that the derailment coefficients derived based on friction formula and creep formula compare fairly well with Nadal's formula at large AoA while

significantly deviating at small to large negative AoA, as shown in Fig. 10.

Zeng et al. [64] additionally propose another derailment criterion with a set of analytical equations. This criterion is a function of friction coefficients, yaw angle, contact angle, normal force, and specific coefficients. For extreme conditions, the study proposes an additional coefficient for assessing an impact-based derailment criterion that is based on the lateral impact load to vertical load ratio. The conclusions derived are close to the previously discussed papers. Zeng and Wu [64] designed a 3 DoF wheelset model. The authors mention that the main parameters are the wheelset lateral displacement, velocity, yaw angle, and rate, as expressed by the following equations:

$$L = -N_L \sin(\delta_L + \Phi) \cos \psi + F_{Ly}, \quad (8)$$

$$V = N_L \cos(\delta_L + \Phi) \cos \psi + F_{Lz}. \quad (9)$$

The authors conduct a thorough simulation study to validate their new criterion. The results demonstrate good prediction capabilities and show that the new criterion can accurately replicate the predictions of derailment coefficients.

Shabana et al. [66] propose a Nadal-like formula based on a basic kinematic approach. A rather large influence in the direction of lateral and vertical forces changes the orientation of the sign convention followed. An assumption of a single DoF wheel climb distance on the rail makes it more conservative than the existing limit. The authors made an

honest effort in this regard and concluded not to recommend it for derailment considerations.

Guan et al. [67] put forward an AoA-based derailment criterion, like that of Nadal's coefficient, but with the inclusion of an equivalent friction coefficient that depends on the AoA. Two creep theories are used to model the new formulation. The authors compare the two through testing. The results show good agreement with the quasi-steady wheel derailment approach taken. Some comparisons of the results with experimental data demonstrate good accuracy, with less than a few percentages error difference. The results for high-friction (i.e., dry) and low-friction (i.e., "oily," wet, etc.) are shown in Fig. 11.

In a more recent study, Kuo and Lin [68] present a comprehensive analysis of Nadal's derailment criteria, which is more detailed than the earlier suggestion, using tangential contact equilibrium. Although complete, the proposed criteria find no difference from Weinstock's [57]. Another in-depth examination of Nadal's derailment coefficient for different states of motion for an object on a slope is provided by Huang et al. [69]. The authors propose a so-called new limit, which closely resembles, and is the same as, what had existed and was referenced in earlier papers. Zeng et al. [70] use a geometrical criterion rather than a force-based one. They test and validate their theory using an IWS on a TTC track. The IWS can accurately determine the contact position and contact forces, thus helping provide L/V forces for different L/V limit criteria. The results show a good approximation of the risk of derailment using the wheel–rail contact position-based approach. Yet, it is only validated for low

speeds, restricted by the testing apparatus. Further investigation into this aspect does not exist.

Similarly, Koo and oh [71] establish a new derailment criterion considering dynamic and geometrical effects of a wheelset. An initial dynamic model is created theoretically to establish the new criteria, considering the normal state condition, wheel slipup, rollover, and other dynamic states during wheel lift. Based on these calculations, new L/V derailment coefficients are curated for both wheel climb and wheel lift phenomena:

$$\frac{L}{V} = \frac{(F - ma_{cx})(1 + \mu\beta) - \mu W_L(1 + \gamma) + \mu\gamma W_R}{(1 + \gamma)W_R - \gamma W_L + \beta(F - ma_{cx})}. \quad (10)$$

Equation (10) includes the lateral force components of the wheelset, the forces at CG, and the friction coefficient (μ), and wheel load at the two wheels (W_L and W_R). The terms γ and β are geometric entities that depend on the wheel configuration and parameters. Based on the above equation, the authors thoroughly analyze mechanical factors such as lateral acceleration, gyro effect, axle bearing positions, track gauges, wheel load variations, and wheel radii. The authors provide a new perspective on the understanding of these mechanical factors. It is recognized that lateral accelerations, although not explicitly included in the equation, play a crucial role as they decrease with an increase in running speed and wheel conicity. The probability of derailment increases with decreasing clearance between the wheel flange and the side of the rail. Such a decrease can happen with gauge narrowing (when track width reduces) or wheel spacing increase

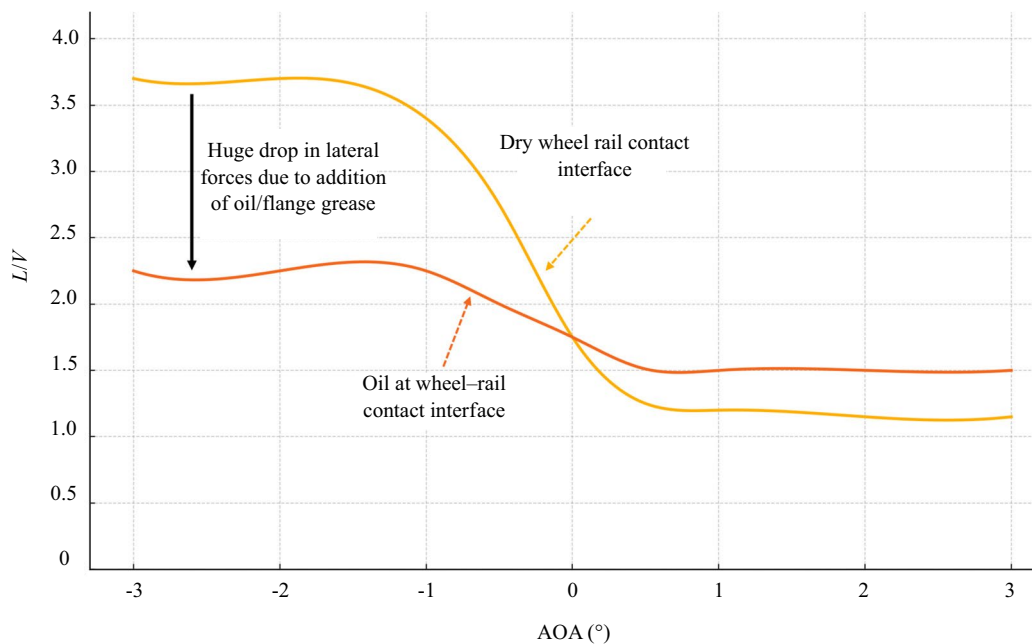


Fig. 11 Derailment coefficient from proposed criterion showing the effects of a contaminated wheel–rail contact (adapted from [67])

(when the distance between the wheel center and the axle bearing support increase). A similar effect is observed when the wheel radius increases, wherein the probability of derailment increases when the contact occurs at a higher rolling radius, such as that observed on the flange.

Shabana et al. [72] introduce a spatial L/V dynamic formulation based on Nadal's limit. No conclusions were drawn that show any progress with this new formulation. The authors provide a variety of reasons to suggest the need for further investigations using tools like multibody system (MBS) simulation models proposed in [73].

Table 2 summarizes various variations established in place of Nadal's derailment criteria, aptly underscoring several dynamic and static parameters added to the established contact force equations. Evaluating past studies suggests that a great deal of work has been done in providing different variations and advanced derailment criteria. Yet, there is a need for more experimental and numerical approaches, which are warranted. Owing to this, the industry still heavily relies on Nadal's derailment criteria to predict the risk of derailment and estimate the effect of track geometry and wheel or rail profile on derailments caused by lateral forces. The significant advances in contact modeling and the development of models that can more accurately predict contact locations, contact angles, and contact forces are expected to provide more precise derailment criteria that improve those suggested to date.

5 Importance of contact forces

This review highlights the history of research conducted on derailment studies due to the wheel and rail interface. Several parameters influence the wheel–rail contact patch, including wheel and rail profiles, wheel and rail wear and defects, third-body layers, contaminations, and most notably, the contact forces (lateral, vertical, and longitudinal) [74]. The contact patch is also influenced by the general direction of lateral and vertical contact forces on the wheels traversing a tangent track, such as shown for curved and tangent track as shown in Fig. 12. The importance of contact forces is evident through the Nadal derailment coefficient in Eq. (2), which indicates that the lateral and vertical forces are intertwined with the flange angle and the coefficient of friction. Considering this as a baseline understanding, several researchers have focused on the derailment coefficient using experimental and simulation environments to correlate with the phenomenon of wheel derailment (wheel climbing).

To provide a more comprehensive discussion, we will divide the influencing factors into two categories: (1) lateral,

vertical, and longitudinal contact forces and (2) wheel and rail design.

5.1 Lateral, vertical, and longitudinal contact forces

The derailment coefficient represents a crucial relationship between lateral and vertical forces and the flange angle between the wheel and the rail, taking into account the coefficient of friction [75]. Although conservative, this relation is fundamental to several railroad researchers and the industry, which works heavily on addressing the issue of wheel derailment due to numerous operational conditions, as mentioned in the previous section. One type of derailment of great interest is wheel climb, also known as flange climb derailment. Various wheel and rail parameters influence the occurrence of this phenomenon, including the AoA, contact angles, friction coefficient, lateral and vertical forces, and wheel and rail profiles. Therefore, it is essential to estimate and measure these influencers, as well as lateral and vertical forces, under various running conditions, which are typically not controlled in the field. To begin with, measuring these contact forces in-field conditions has always been challenging.

5.1.1 Experimental approaches to measure global forces resulting in derailment

Brocciolini et al. [76] highlight the lack of accuracy in measuring lateral and vertical forces while a train is in motion on the track. They highlight how most research conducted before 2014 used electric resistance strain gauges attached in the normal direction to specific locations on the rail. These strain gauges are used to measure the complex deformation loads to calculate the forces. The authors introduce a wayside measurement device that can measure lateral and vertical forces exerted by a train passing at low speeds on a flat curve. They use a comprehensive FE model analysis to conduct, calibrate, and compare the performance of the wayside sensor package, which features two strain gauges attached to the rail web to measure vertical forces. They additionally discuss different methods to test the accuracy of lateral forces. Although a conventional strain gauge-based system is used, the lifespan of such devices has always been short, owing to several shortcomings and harsh environmental conditions. An illustration of a conventional strain gauge design employed by the industry is shown in Fig. 13.

Matsumoto et al. [77] have attempted to address this issue by developing a measuring device capable of measuring all contact forces, thereby determining the derailment coefficient for a wide variety of curves on commercial lines. A device was proposed that does not employ strain gauges but instead uses the axle spring and bending deflection of the wheel, using non-contact gap sensors (Fig. 14). The

Table 2 Summarized variations to Nadal's derailment criteria using numerical and analytical methods

S. no.	Author (year)	Key parameters/modifications	Validation method	Applicability and key insights	Remarks
1	Nadal (1896) [49]	–	Analytical	Fundamental L/V limit for flange climb; baseline for safety assessments	Over-conservative; excludes AoA, contact angle, and dynamic effects
2	Spinola Barbosa (2004) [61]	ψ (yaw angle), δ (contact angle), k_x, k_y	Analytical/theoretical	Integrates yaw and contact geometry into a 3D rolling model	Complex; lacks experimental support
3	Shu et al. (2005) [62]	D (flange climb distance), ψ	Limited experimental	Bi-parameter distance model for flange climb; valid for higher ψ (> 5 mrad)	Restricted to AAR-1B and AREMA-136 profiles
4	Zeng and Guan (2008) [63]	δ, ψ, μ [as a function $C(\psi)$]	Analytical	Simplified quasi-static relation; less conservative than Nadal for moderate ψ	Needs further field verification; deviates at small AoA
5	Zeng and Wu (2008) [64]	ψ, δ, μ , impact ratio	Simulation	Considers yaw, contact angle, and lateral impact; applicable to transient derailments	Theoretical formulation only
6	Durali et al. (2010) [65]	y (lateral disp.), y' (lateral vel.), ψ , yaw rate	Dynamic simulation (3 DoFs)	Captures transient L/V and stability; applicable to dynamic wheelsets	Sensitive to parameter tuning; no field validation
7	Guan et al. (2014) [67]	μ_e (equivalent friction), ψ, F_x, F_L	Simulation and laboratory tests	Adhesion variation modeling; good correlation with measured derailment limits	Small-scale validation only
8	Koo et al. (2014) [72]	W_L, W_R (wheel loads), β, γ (geometry coefficients), a_x (lat. acc.), μ	Analytical/parametric	Combines wheel lift and flange climb phenomena; includes geometric asymmetry	Complex implementation; high input requirement
9	Huang (2021) [70]	θ_s (slope angle), frictional components of lateral slide	Theoretical	Re-interprets Nadal's limit under dual derailment modes; geometric clarity	Conceptual; unvalidated
10	Zeng et al. (2022) [72]	x_p (contact position), ψ, F_L, F_V	Experimental	Geometric criterion linking contact shift to L/V ; effective for low-speed operation	Tested only at low speeds; limited scalability
11	Shabana and Ling (2023) [73, 74]	$\psi, \delta, 3D$ spatial motion variables (x, y, z), Φ (rotation)	Analytical	Spatial dynamic L/V incorporating full 3D motion	Requires numerical validation

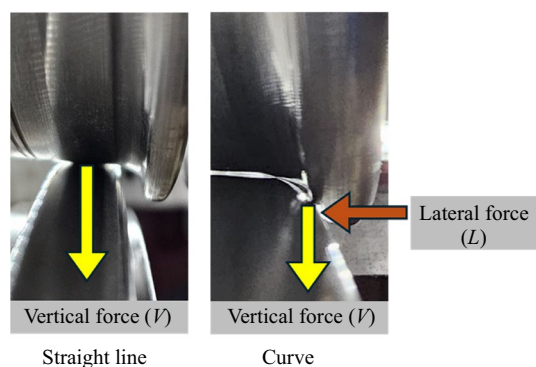


Fig. 12 Contact forces generally observed in a tangent and curved tracks

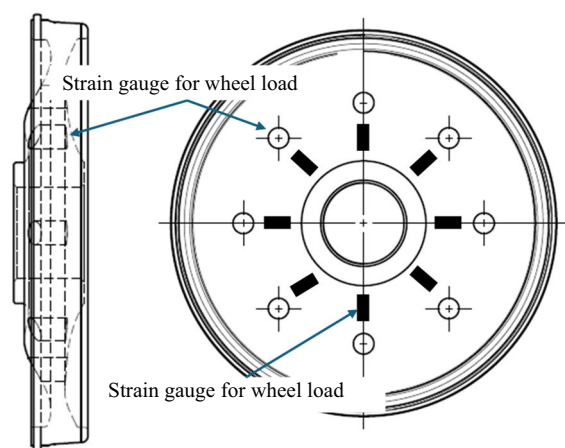


Fig. 13 Conventional instrumented wheel design with locations of strain gauge mounted on the web of the wheel

technique adopted is the use of a testing bogie, which was also trialed on a revenue service line for continuous measurements. They highlight how the derailment coefficients increase as the coefficient of friction increases. The device can also account for inevitable fluctuations in the coefficient due to track irregularities and vibrations at each curve. However, the study remains limited by its dependence on idealized track and bogie conditions, and its validation scope is confined to moderate-speed runs. Additional multibody dynamic simulations were performed using the FASTSIM algorithm for wheel–rail contact, for comparative purposes, which showed a decent level of consensus. Moreover, the work primarily establishes a correlation between friction level and derailment coefficient rather than causal quantification of dynamic instability. The importance of the coefficient of friction in predicting derailment coefficients is also highlighted by Bocciolini et al. and Matsumoto et al. [78, 79]. Hence, while Matsumoto's approach represents a meaningful

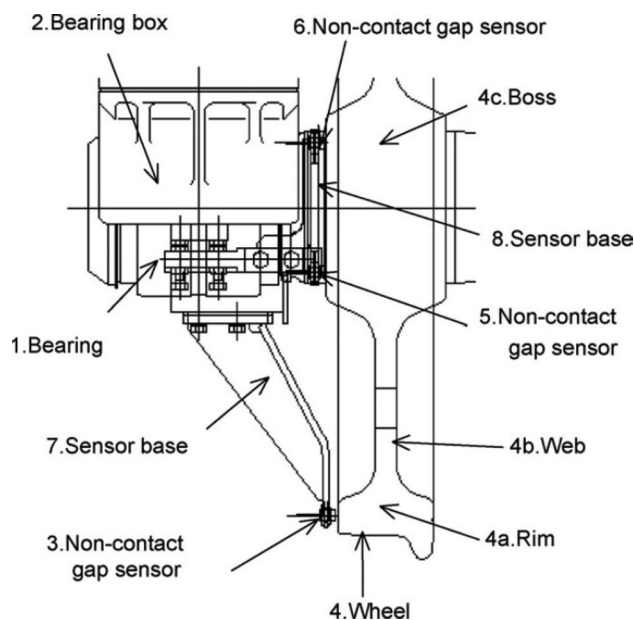


Fig. 14 Non-contact gap sensors used by Matsumoto et al. [79] (reproduced by permission from the publisher)

step toward continuous in-service derailment monitoring, its scalability and robustness under high-speed or irregular track conditions remain open for further investigation.

Cortis et al. [80] propose another unique strain gauge sensor that can isolate the lateral forces from vertical forces using the Sistema di Misura dei Carichi Trasversali (SMCT) system. The wayside device comprises four strain gauges capable of measuring eight grids at a 45° angle. These gauges are glued on the upper surface of the rail foot and placed symmetrically to the middle axis of the span. The sensors are also placed at a specific distance from each other based on where the optimized linearity is observed between the sensors. The sensor configuration was tested under laboratory conditions, where the advantage of the Wheatstone bridge layout leads to the elimination of strains due to vertical forces. An adaptation of the author's design is illustrated in Fig. 15a. This produces an output that is directly proportional to the lateral forces. After the test setup was analyzed in the laboratory for proof of concept, the optimal distance between the sensors was set to 200 mm. The authors then simulate the results using a 3D FE model with desired SMCT rig measurement characteristics to validate the experimental results. Further testing was conducted in the field, where substantial lateral force measurements were recorded during the beginning and end of the curved track. These results substantially help in understanding the curving dynamics, and this prediction enables curve negotiations when different types of trains pass by. The derailment coefficient was also recorded. While the results look promising,

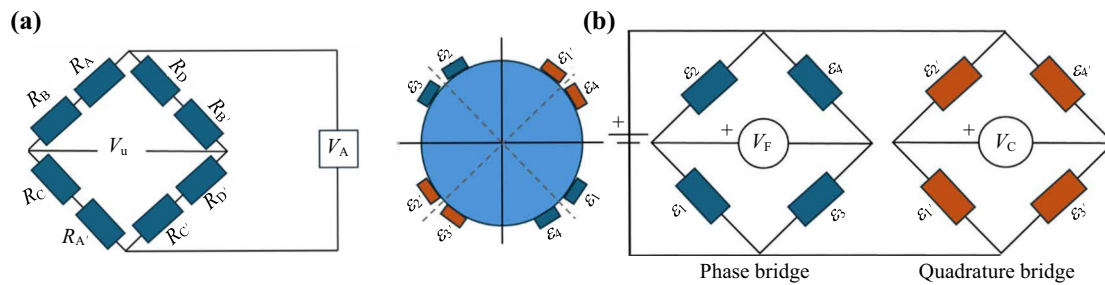


Fig. 15 a, b Wheatstone bridge designs generally used in different output requirements (adapted from [80, 81])

the industry has yet to employ such a device in full-scale revenue service requirements. The sensors rely on adhesive bonding, which may degrade under temperature variation and ballast vibration, potentially leading to calibration drift over long-term deployment. Moreover, the system was only evaluated on limited track segments and did not assess high-speed or mixed traffic environments where signal noise and wheel profile variability could affect accuracy. The absence of a continuous data-retrieval protocol also constrains its scalability for network-wide monitoring. Another standard Wheatstone bridge design is shown in Fig. 15b.

Now, the challenge of such an approach is to retrieve the recorded data using complex data-acquisition (DAQ) systems that require an operator to access the device on-site. This is not only safe but also time-consuming, considering the number of such devices necessary to gather data from all over various tracks. This questions the possibility of a setup, such as a DAQ retrieval system, which can provide continuous measurements that can be observed in situ (at a particular moment). Tanimoto et al. [82] attempt to answer this by using a conventional strain gauge system to measure the reaction forces acting both laterally and vertically. The data collected by the DAQ are automatically transferred to the nearest command center, as they call it, using optical fibers. This process helps predict track maintenance requirements by analyzing the results from the device, including the derailment coefficient, and deviations from the ideal or historical measurements. This short-cycle feedback helps prevent unforeseen derailment situations. In this regard, several research attempts have been tested experimentally. The method's calibration stability under vibration, temperature fluctuation, and ballast movement remains uncertain. The most common method for measuring contact forces on the field has been via an instrumented wheelset (Magel et al. [83]) or strain gauges mounted on the rail (instrumented rail).

Wu et al. [84] aptly conclude that an optimized strain gauge bridge design is needed to improve both test accuracy and reduce the influence of environmental or

other external noise. Based on these results, one can identify the potential drawbacks associated with the scope of the device. The majority of these designs involve physical modification to both the wheel axle (including the wheel) and the rail; no appropriate testing is done on the result of the longevity of the device itself as the strain gauges are usually where malfunctions are observed and the structure where the sensors are implanted as they bear extensive loads thus could lead to faster wear and tear causing calibration issues [85]. This requires additional thought and consideration of alternative methods to measure contact forces on the field.

Table 3 summarizes the experimental measurement techniques used to record vertical and lateral contact forces, thereby increasing understanding of derailment forces. The collective efforts observed until now have substantially advanced how researchers measure the contact forces that lead to derailment. Over the years, experimental approaches have evolved from using traditional strain gauges to sophisticated non-contact and IWS designs, improving the precision and repeatability of lateral and vertical force measurements. Yet, these techniques are not without challenges. Strain gauge drift, temperature effects, short sensor and slip ring lifespan, and high installation costs still limit long-term field use. While non-contact and hybrid sensors offer better durability, they demand complex calibration and have yet to see consistent validation when used in-field conditions. Moreover, most experiments are confined to controlled or low-speed environments, making it difficult to directly translate the findings to heavy-haul or high-speed test conditions. Together, these shortcomings reveal a critical gap between laboratory precision and field practicality. Bridging this divide will require rugged, self-calibrating sensing systems and standardized testing procedures that can sustain accuracy under the dynamic and unpredictable conditions of railway operation. Building on these experimental insights, several researchers have turned to advanced simulation models to complement field data and explore derailment

Table 3 Summarized reviews on experimental approaches to measure global forces resulting in derailment

S no	Author (year)	Technique used	Scope	Remarks
1	Bocciolini et al. (2014) [77]	FE simulation and wayside strain gauge device on rail web	Measured lateral and vertical forces on the track	Effective for low-speed tests and sensor calibration but has a short lifespan under field conditions; requires frequent maintenance and recalibration
2	Matsumoto et al. (2006, 2008, 2012) [78–80]	Non-contact gap sensors and an instrumented test bogie supported by FASTSIM simulation	Measured all contact forces and continuous derailment coefficient (L/V) on curved track	Demonstrated field-worthy monitoring method for commercial lines; accuracy depends on friction coefficient and track irregularities; good potential for in-service applications
3	Cortis et al. (2020) [81]	Experimental with a strain gauge system and 3D FE simulation	Measured lateral forces and derived L/V ratio through laboratory and field validation	High accuracy for isolating lateral loads; DAQ retrieval is still manual and labor-intensive; limited field deployment so far
4	Tanimoto et al. (2017) [85, 86]	Conventional strain gauges with fiber-optic data transfer for remote monitoring	Measured lateral and vertical forces; real-time computation of derailment coefficient	Enables continuous health monitoring and maintenance prediction; high practical applicability, but costly installation and reliance on a delicate fiber network

behavior under a wider range of operating and geometric conditions.

5.1.2 Simulation and numerical environment to evaluate derailment

The numerical and simulation techniques used to predict the effect of various parameters on derailment in various conditions are summarized in Table 4. It is important to note that significant progress has been made over the years in the field of simulation models and the technologies that arise from them. This advancement has led to advanced, faster, and more accurate models.

An example of such an advancement is the use of a large DoF dynamic model by Zhu et al. [86] using the SIMPACK tool. All the forces exerted by the axle box were taken as a response to the mathematical model. The results of the model were further inverted to give the wheel–rail forces. The results compared reasonably well to the models that have previously existed. Using force and moment equilibrium equations, however, the study’s simplifications, such as assuming rigid wheelsets and neglecting rolling dynamics, limit its applicability under real-world transient conditions. Further experimental validation and inclusion of suspension dynamics are necessary to enhance the model’s predictive reliability for derailment assessment.

Wei et al. [87] used an indirect method to measure wheel–rail forces. Forces and moment equilibrium equations were used. While comparing results from SIMPACK, good accuracy was maintained, keeping the error under 15%. It is essential to note that this model has its limitations due to certain assumptions made. The rolling motion of the wheelset is ignored, and the wheelset is treated as a rigid body. A primary spring for the wheelset is an essential consideration for this model. More laboratory tests and additional field conditions are necessary to improve the predicted output.

Over the years, there have been significant advancements in methods and simulation models for measuring or predicting wheel–rail contact forces, particularly vertical and lateral forces. One must use this to their advantage and extend these models to study and predict derailments. This will provide better validations and cater to practical needs, as this is the need of the hour.

Creep forces, or longitudinal forces that arise from the relative slip between the wheel and the rail, also play a significant role. Santamaria et al. [50] attempt to summarize recent advances and establish a new three-dimensional approach for creep forces. The authors use this proposed approach to predict derailment limits for small and large AoA. It is well known that Nadal’s derailment limit is valid primarily for large AoA, making it conservative due to its assumptions; the researchers compare their results to predictions from Nadal’s limit. The authors do extensive

Table 4 Summarized reviews on simulation and numerical environment to evaluate derailment

S. no.	Author (year)	Technique used	Scope	Remarks
1	Zhu et al. (2013) [84]	Simulation only; inverse dynamics approach using SIMPACK	Derived wheel–rail forces from axle box responses to estimate lateral and vertical loads	Accurate within ~ 15% for static and low-speed conditions; neglects wheelset rolling motion; requires further laboratory and field validation for practical use
2	Wei et al. (2014) [88]	Indirect method for force measurement and simulation	Estimated wheel–rail forces via force and moment equilibrium equations; benchmarked with SIMPACK	Maintains good accuracy but assumes a rigid wheelset and a simplified spring model; further testing under dynamic loading is needed for industry application
3	Santamaria et al. (2009) [50]	Analytical and Simulation based on 3D creep force formulation	Developed a three-dimensional creep force model to predict derailment limits under varying AoA and contact geometry; compared results with Nadal's limit	Captures the interaction between longitudinal and lateral creepages, demonstrating that reduced normal load lowers the derailment threshold; useful for theoretical understanding but requires experimental validation
4	Barbosa (2009) [89]	Analytical 3D simulation based on Kalker linear theory and FASTSIM algorithm	Modeled flange climb under increasing lateral force; studied the relation between longitudinal and lateral creep forces	Captures force transition post-flanging; Nadal's limit remains conservative; valuable for model-based risk prediction but could improve with vehicle suspension consideration
5	Molatefi and Mazraeh (2016) [91]	Analytical and numerical based on a 5-DoF single-wheelset Euler beam model	Studied flange climb mechanism and the effect of creep forces and lubrication on L/V behavior	Identifies L/V rise until derailment, then drop; recommends lubrication for safety but notes traction loss trade-off; useful for controlled testing scenarios and to understand wheel sliding
6	Shu et al. (2020) [92]	Experimental on RCFS and simulation	Measured creep curves under dry and wet conditions; validated simulation for low-speed wheel climb	Demonstrates friction variation's impact on derailment coefficient; valuable for low-speed and turnout safety assessment; requires extension to worn wheels and high-speed cases

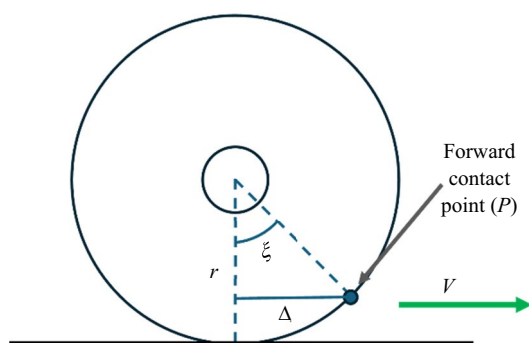


Fig. 16 Forward contact point in the diametral plane. A new approach to understand complex contact conditions in a curve (adapted from [50])

theoretical calculations to analyze the creep force in a 3D system. As the flange comes into contact, the point is longitudinal forward. They took this into account and referred to it as “P.” This point “P” is at an angle denoted by ξ from the vertical diametral plane of the wheel, as shown in Fig. 16.

A complex matrix transformation retrieves the new expressions for vertical and lateral components. These forces are observed to be a function of the AoA (ψ), forward contact angle (ξ), wheel–rail contact angle (δ), and tangential force components (t_x and t_y). For ease of calculation, several assumptions are considered, and a new, updated form of the derailment coefficient is achieved. Equation (11) provides the derailment coefficient formulated by the researchers.

$$\frac{L}{V} = \frac{\tan \delta \cos \psi - t_y \mu \cos \psi - \frac{t_x \mu \sin \psi}{\cos \delta}}{t_y \mu \tan \delta + 1}. \quad (11)$$

Based on this, several iterative, parametric theoretical calculations, including some simulations, were done. The authors aim to provide a model that is computationally fast and provides realistic values. Based on the observed results, the authors conclude that the influence of the yaw angle and forward contact angle can be disregarded, as these values do not significantly impact the results. A significant divergence of derailment limit is observed when the normal load has been reduced, i.e., a reduction in the expected load results in a considerable drop in the derailment limit initially provided by Nadal’s coefficient. However, this approach is highly model driven, as the exact identification of point P and its angular position (ξ) cannot be reliably measured in real-world conditions. Nonetheless, it presents a valuable framework for future simulation studies, where such idealized geometric definitions can be leveraged to refine derailment prediction models under controlled virtual environments. Barbosa et al. [88] studied this complex dynamic phenomenon using an analytical approach. The influence of longitudinal force and the effect of spin creepage are considered in a

3D analytical space. The results demonstrate the significant impact of contact forces after flanging. Due to the wheel conicity, a distinct pattern emerges in the behavior of transverse and normal forces. They compare the results to the FASTSIM algorithm [89]. In their case, Nadal’s derailment limit remains conservative when comparing the results of L/V to friction coefficients; adaptation of the results is shown in Fig. 17. The authors iterate how the large wheelset AoA increases tangential saturation. They additionally provide an interesting correlation between longitudinal and lateral creepage (due to AoA). They demonstrate how a longitudinal creepage of 1% yields a lateral force of 70%, a value typically observed when the lateral creepage is 1%. When the longitudinal creepage becomes 2%, the lateral force drops to 50%. This single-wheelset testing showed the L/V limit to be very close to that predicted by Nadal. They conclude that it is essential to understand the suspension system further, to account for their effect in derailments, since a rigid suspension may yield results that differ from field occurrences.

Molatefi et al. [90] developed a model with a five-DoF single wheelset of Euler beams. Lateral displacements are introduced into the system through impulsive loads. The AoA is maintained at a constant value. For this research, several wheel–rail contact algorithms were used. The results indicate a rise in the L/V ratio until derailment, after which there is a significant drop in L/V , as expected. They conclude that the FASTSIM algorithm consistently yields accurate results. The authors discuss how the wheel velocity and the system’s primary suspension do not affect the maximum L/V recorded. The authors provide a means to avoid derailment, mentioning that lubrication of railheads could improve running safety. The impulsive loading approach also does not encompass the continuous track–vehicle interactions or variable adhesion states occurring in revenue service. Future studies could aim to fill this gap by integrating non-linear suspension behavior, variable creepage conditions, and experimental validation using IWSs to better correlate the L/V evolution predicted by the model with measured derailment tendencies in dynamic environments. This is important because the risk of derailment increases due to significant differences in the friction coefficient between the two rails. On the other hand, grease/lubrication contamination results in a significant drop in tractive forces that could result in wheel sliding, requiring more motive effort or other traction enhancers to move the wheel forward and prevent derailment.

Shu et al. [91], from the Transportation Technology Center, used both new and used Class C wheelsets on the rolling contact fatigue testing rig (RCFS) to measure and evaluate wheel–rail contact friction, thereby assessing creep curves under dry and wet rail conditions. The authors demonstrate that any variation in the coefficient of friction will have a detrimental effect on flange climb derailment.

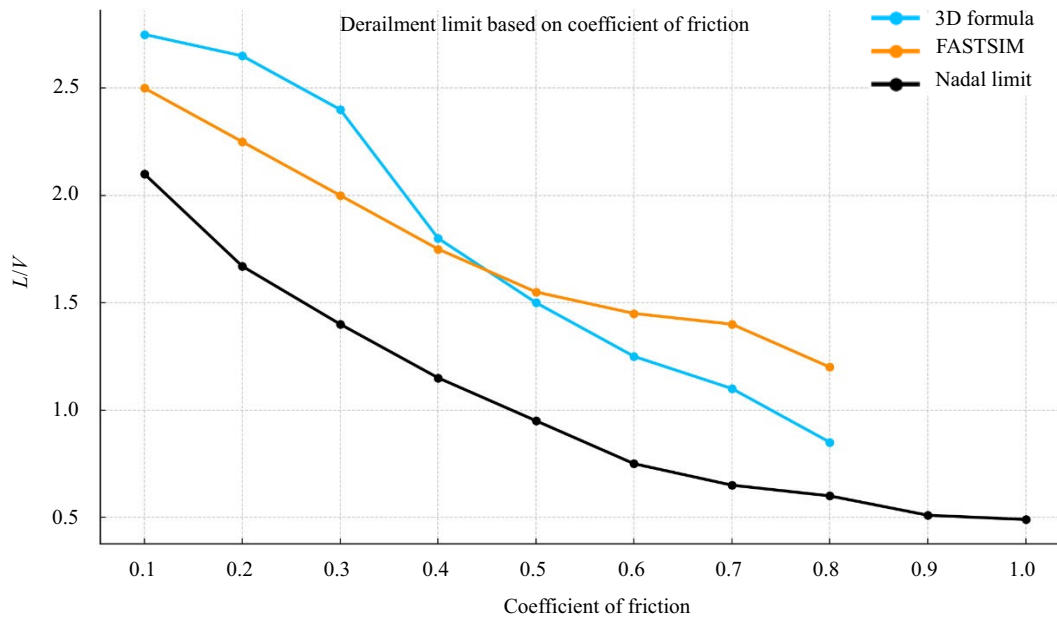


Fig. 17 Results of derailment limit from new 3D formula show a nonlinear relation of L/V and the coefficient of friction (adapted from [61])

They also iterate that this is true when new wheels are installed. No conclusions were drawn when the wheel profile was worn. An additional NUCARS flange climb simulation was conducted under similar operating conditions, using the updated creep curve model derived from the RCFS experiments. The simulations were performed for a number 8 turnout. With the vehicle speed of 24 km/h and Hertzian contact stresses, the derailment coefficient was predicted to exceed the limit when the coefficient of friction was 0.5, and the flange angle was 63° . With this, recent research efforts to measure/evaluate/predict contact forces have been concluded on how to assess derailment types using such approaches.

The studies mentioned above demonstrate major progress in using computational models to simulate derailment behavior, offering valuable insights that are often difficult to obtain experimentally. These efforts have applied inverse dynamics, analytical formulations, and multi-degree-of-freedom models to reconstruct wheel–rail forces and replicate complex contact scenarios. However, most of these approaches rely on iterative or quasi-static simulations rather than continuous dynamic modeling. As a result, transient effects—such as rapid load transfer, vibration coupling, or wheelset relaxation—are often simplified or completely neglected. This limits their ability to capture the short-duration instabilities that typically precede derailment. Moreover, while several of these models show good correlation with experimental data, the validations are usually based on isolated laboratory cases rather than extended field datasets, reducing their general applicability. Many simplifications, such as assuming rigid

wheelsets, linearized contact geometry, or constant friction coefficients, further constrain model accuracy. Despite these approximations, simulation tools remain essential for identifying derailment trends and guiding experimental design. It is recommended that future research focus on coupling high-fidelity transient models with validated field data to close the gap between predictive simulation and real-world derailment behavior.

These numerical advancements, though powerful, highlight that derailment is not purely a computational problem but a complex physical event. To truly understand its nuances, such as flange climb initiation, multi-point contact, and dynamic force transfer, experimental and numerical approaches must work hand in hand.

5.1.3 Experimental and simulation (hybrid) studies focusing on flange climb derailment

The contact point, yaw, wheel/rail profiles, contact angles, and derailment coefficient highly influence the condition of wheel climb or flange climb. Wheel climbing occurs in two stages: when a wheel negotiates a curve with an attack angle. In the first stage, as the wheel shifts toward the flange, there exist two points of contact, one on the tread and the other on the flange. A prolonged lateral shift converts this two-point contact into a single-point flange contact point, typically at a higher angle. The second stage is a derailment; the wheel climbs and thus derails. This occurs due to large lateral forces, usually observed by large derailment criteria values [92].

The experimental and simulation (hybrid) studies to better understand the wheel climb derailment mechanism are summarized in Table 5. Researchers have attempted to study this complex phenomenon both experimentally and in an advanced simulation environment. Magel et al. [83] use data from instrumented wheelsets attached to the Amtrak Acela trainsets. The power car (locomotive) and coach cars are compared for tractive forces, L/V ratio, and wear. The investigators emphasize the importance of minimizing high lateral forces, particularly during high-speed train operation. The L/V ratios observed are almost the same for power cars and coaches, but only for the high rail contact point. The low rail observed significant variations in a smaller curve. In a tighter curve, the L/V measured values are much closer. This is attributed to the larger longitudinal tractive forces generated by the power car when compared to the coach. The authors follow the Weinstock criteria and consider wheel climb only if the index is ≥ 1 . In the case of this study, the Weinstock criteria were < 0.4 . This implies that Nadal's L/V is at 40% of the estimated limit to observe wheel climbing. Only the leading axle of the power car exhibited higher chances of wheel climbing. A group of researchers from the safety research laboratory investigated the science behind wheel climb due to low speed [93]. One track is selected for testing with a transition curve of 100 and 110 m (Fig. 18). A test car was selected, previously used in revenue service, a standard commuter train with a bolsterless bogie was used by the researchers. A specific time frame was selected for these tests. Three different measurement parameters were considered. The wheelset was instrumented to provide vertical, lateral, and longitudinal creep forces, including the AoA from the two axles, and the climb distance. Other environmental conditions, like ambient temperature, were measured using wayside devices. Theoretical calculations were initially done, where an equivalent friction coefficient term was introduced, where the coefficient of friction was a function of the AoA.

$$\mu_e = \mu \cdot \frac{\psi K_{22}/N}{\left[\mu^\beta + \left(\frac{\psi K_{22}}{N} \right)^\beta \right]^{1/\beta}}, \quad (12)$$

where μ represents the friction coefficient of the outer flange, μ_e is the equivalent friction coefficient, ψ is the AoA, N is the normal load, K_{22} is the lateral creep coefficient, and β is the index of saturation.

To validate this, the authors compared the equivalent coefficient of friction values due to onboard measurement devices to those of the AoA. The results indicate how the measured L/V is more severe than the conventional guide of 0.95 when the curve is sharp, i.e., < 150 m curve radius. However, the equivalent formulation works well with lower AoA and deviates when higher speeds are considered. More

research effort is warranted to replicate this experimental setup for stronger comparative assessment. As, the use of a single test vehicle and specific track geometry limits generalization across different bogie designs and rail profiles. Future studies should test with different speeds, larger curves, and more types of vehicles, while using faster data collection tools to better capture quick changes that happen during wheel climbing.

In a similar attempt to understand the wheel climb phenomenon at low speeds, Ishida et al. [94] assess the safety aspect of flange climbing. A safety assessment parameter and a safety index are introduced. The estimated L/V ratio is based on a critical value limit proposed by Nadal. Based on the estimation design, one can consider it as a safety factor [Eq. (13)]. Some laboratory measurements examined the wheel climb distance on a roller rig. The roller diameter of 1600 mm is used on a full-scale wheelset. Field tests were also conducted to measure creep forces on a rail track. The Railway Technology Research Institute (RTRI) line was employed for this effort. The rail profile was the Japanese 50 N, and a standard wheel profile with a 60° flange angle was used. Wheel climb was observed at the center of the track curve. These measurements were analyzed to study the influence of AoA on creep forces. The measurements from the laboratory setup showed good consensus with the existing knowledge of the effects of AoA on longitudinal and lateral coefficients. The field test values observed were very close to the theorized values, i.e., the estimated model of L/V . A new safety index is introduced, independent of the rail and wheel friction coefficients. Based on these indices under the low-speed conditions, the outside wheel traversed a distance of 1 m from the beginning of the flange climb to derailments. Based on further analysis, a standardized procedure for safety assessment under low-speed operational conditions is proposed.

$$\text{Estimated } \frac{L}{V} \text{ ratio} = \frac{\text{Critical } L/V}{\text{Estimated } L/V}. \quad (13)$$

They iterate that the train still runs safely if the measured L/V ratio is less than a 5-Hz limit frequency index. Another scenario exists if the L/V exceeds the conventional L/V (Nadal's) limit; an off-loading factor $\Delta Q/Q_0$ should be measured, where ΔQ is a change in wheel load observed, while Q_0 is the static wheel load recorded. The train operation is considered safe if this ratio has a frequency of < 5 Hz. $\Delta Q/Q_0$ safe operation zone is < 0.4 . This safety assessment can test other operational conditions at low speeds to test the accuracy of the curated safety criteria. Although this is an appreciable effort, it will be challenging to use as it must be further assessed in other running conditions. Having a wheelset train to measure forces is also challenging as it could cause

Table 5 Summarized review on experimental and simulation studies focusing on flange climb derailment

S. no.	Author (year)	Technique Used	Scope	Remarks
1	Magel et al. (2008) [84]	Experimental (instrumented wheelsets on Amtrak Acela)	Measured longitudinal, lateral, and vertical forces under high-speed operation; compared L/V ratios for power and coach cars	Demonstrated how high lateral forces influence wheel climb; verified Nadal's and Weinstock limits; highly relevant for high-speed rail safety monitoring
2	Iijima et al. (2009) [94]	Experimental and analytical	Tested wheel climb behavior on transition curves; introduced equivalent friction coefficient as a function of AoA	Valid for small to moderate AoA; confirms higher L/V in tight curves; requires broader validation for higher speeds and multiple track conditions
3	Ishida et al. (2006) [95]	Experimental (full-scale wheelset and field test)	Developed a safety index for low-speed flange climb using measured creep forces and climb distances	Provided a quantitative wheel climb safety assessment; reliable for low-speed operations but difficult to generalize to different bogies or rail profiles
4	Nagumo et al. (2008) [96]	Experimental with a 1/5-scale model wheelset and analytical	Examined wheel climb under unbalanced load using optical sensors; evaluated yaw angle and force influence	Well-controlled parametric testing; applicable to small AoA and low speeds; scaling effects limit direct field translation
5	Nagase et al. (2008) [97]	Experimental with a 1/5-scale bogie	Examined wheel climb under unbalanced load using optical sensors; evaluated yaw angle and force influence	Well-controlled parametric testing; applicable to small AoA and low speeds; scaling effects limit direct field translation
5	Braghin et al. (2006) [98]	Experimental with a BU300 full-scale roller rig and simulation	Studied quasi-steady flange climb and proposed "σ-limit" and "saturation-limit" criteria	Solid experimental and numerical correlation; quasi-static assumption limits dynamic reality; valuable benchmark for later simulation validation
6	Matej [100]	Experimental (controlled wheel flange climb on curved/straight track)	Measured wheelset response during forced flange climb events	Demonstrated controlled test feasibility; provides validation data but lacks detailed methodology for replication. Note: not a peer-reviewed article
7	O'Shea and Shabana (2017) [102, 103]	Simulation and analytical	Investigated wheel climb initiation with geometric distinction between flange and contact angles	Identified need for AoA-based geometric consideration; questions the conservatism of Nadal's limit; theoretical but impactful for future derailment modeling

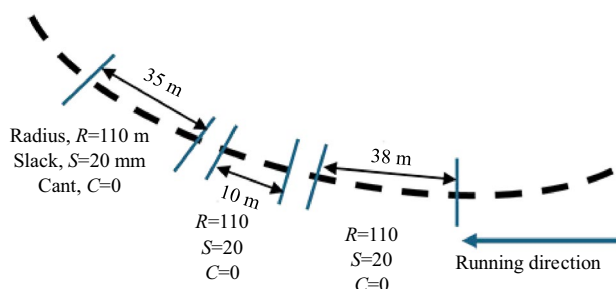


Fig. 18 Curved track replication used in testing with different curve radius (adapted from [93])

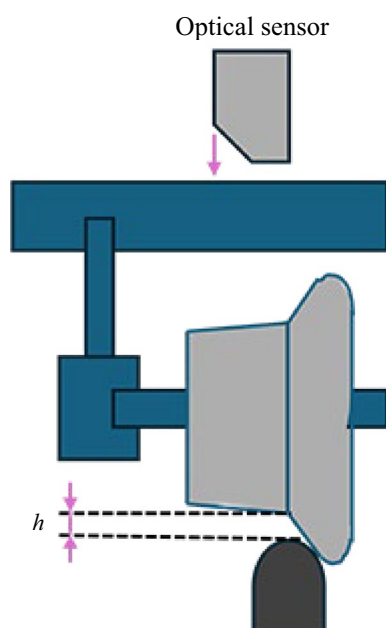


Fig. 19 Optical sensor position and wheel climb measurement for accurate measurement of wheel climb mechanism (adapted from [94])

severe financial requirements to the industry. Nagumo et al. [95] employ a 1/5-scale model wheelset to study the wheel climb under a low-speed, unbalanced load condition, using an optical sensor as illustrated in Fig. 19. Four parametric test cases are performed for a vehicle speed of 10 km/h. A dynamic lateral load is simulated at a rate of 50 N/s. As the tests are in the time domain, the climb height (h) is measured until derailment. The derailment criteria are set to be 6 mm wheel climb height. The duration until derailment is inversely proportional to the yaw angle. The authors also observe a significant impact of yaw angle, unbalance ratio, and dynamic changes in external forces on the onset of wheel climb derailment. Basic numerical analysis was performed for the transition from two-point contact to single-point flange contact. An analytical method is employed to compare the experimental setup. While it is well suited for small AoA, further work is required to provide a better quantitative agreement. Yet again, such studies need different scenario such as high speeds, other curved tracks to improve the dynamic effects into the proposed limits. As such technology requires wide variety of testing when the plan is to make it more practically suited for field base systems.

In the case of Nagase et al. [96], a 1/5th scaled bogie (Fig. 20b) is used to evaluate wheel climb and derailment coefficient. The bogie has a wheelbase of 2100 mm and a wheel diameter of 172 mm. The flange angle in the scaled bogie is 60° . A laser sensor distance-measuring device records the vertical displacement (wheel climb). The track has a 100 m curve radius. Typical strain gauges were used to measure lateral and vertical forces. Lateral forces were applied due to insufficiency in natural centrifugal effects. With an AoA of 0.7° and specific testing conditions, wheel climb was observed at a lateral force of 390 N, while the static vertical force of 245 N was maintained. This results in an L/V ratio of 1.6. The plots of the adapted results are shown in Fig. 20a. Within a travel distance of 140 mm, the

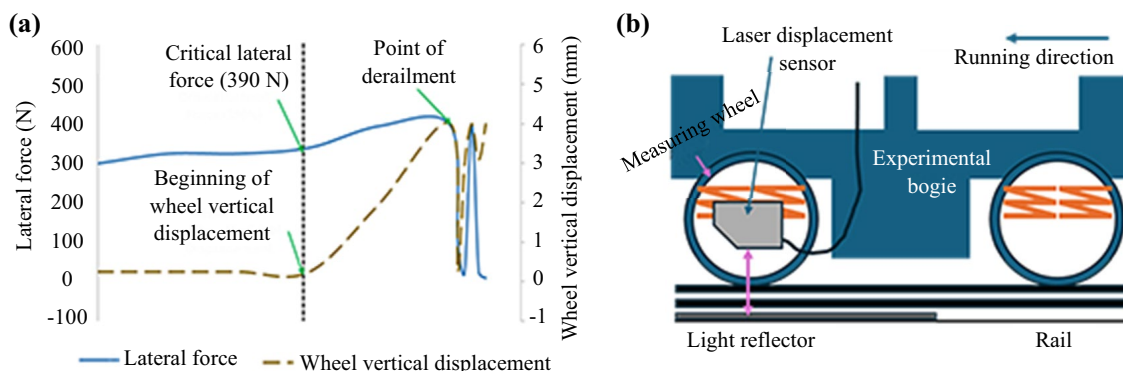


Fig. 20 **a** Experimental results for derailment study done on the test bogie showing exact instance of derailment; **b** experimental bogie with instrumentation (adapted from [96])

bogie derailed. To further understand the influence of the coefficient of friction, two additional tests were performed, one on an oil-contaminated railhead and the other on a worn and sanded rail profile to simulate a rough profile. A significant reduction in wheel climb was observed with oil-contaminated rail, whereas the rough surface condition accelerated wheel climb. The concept of wheel slip-down occurrence is introduced, which refers to the scenario where the wheel falls back onto the rail head due to other factors. The authors have observed this phenomenon and explain that it is due to insufficient adhesive forces as the wheel climbs to counter the effects of gravity, resulting in the wheel falling back down onto the rail. There, the author concludes that the wheel climb derailment does not occur until the wheel climb exceeds the wheel slip down. Speed is also an influential criterion in this regard.

Now, several authors have attempted to provide updated derailment criteria, which aim to substitute the age-old conservative relation proposed by Nadal. Braghin et al. [97] propose two new test criteria, which they later validate using a numerical simulation approach. A full-scale wheelset testing rig, the BU300 roller rig, is used under a quasi-steady flange climb process, as observed in Fig. 20b. A thorough investigation of the experimental results is done to study the effects of the AoA from which the “ σ limit” criterion is proposed. In addition to this, a “Saturation limit” criteria are proposed based on the longitudinal creep forces. The “ σ limit” criteria begin by removing the assumption that normalized tangential forces, typically represented by T/N or F_s , are approximately equal to the coefficient of friction (μ). In addition, the forward flange contact point is considered using a set of linear equations. Considering these tests are under a quasi-static environment, an estimation of F_s is obtained. They additionally use Kalker’s linear theory [42] to set the thresholds. A comparison with Elkin and Wu’s criterion [98] shows a fair agreement with the proposed one. Similarly, the authors derive an equation relating F_s – μ and longitudinal forces. Simulations were done using a numerical model of parameters using the BU300 rig. Additional descriptions of the model can be found in other publications, which are not relevant to the current scope of this review. Although the model demonstrates a good understanding of the experimental results, a more thorough analysis is necessary to overcome the limitations of the quasi-static approach and provide results that are closer to field reality. The experiments were conducted under quasi-static conditions, which do not fully represent the transient dynamics, vibration effects, or time-dependent force variations that occur during real train operations. The model assumes steady contact and constant friction, neglecting the influence of varying adhesion, wheelset yaw, and suspension compliance that can significantly

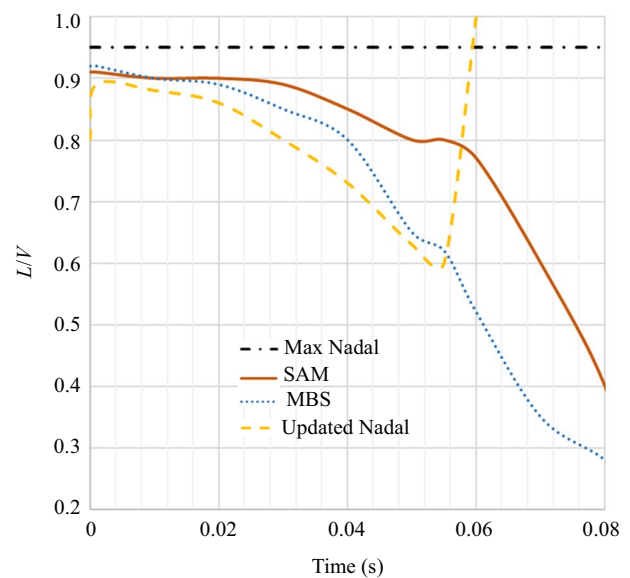


Fig. 21 Measured L/V for different derailment criteria showing complex interplay of several models (adapted from [100])

affect flange climb behavior. Future research could extend these criteria to dynamic testing environments, include variable friction and loading cases, and validate them with IWSs or field data to confirm their robustness under realistic running conditions.

There appears to be a general trend in research that uses advanced testing approaches, such as the use of novel testing rigs, including both full-scale and scaled models, with IWSs [99]. The researchers additionally proposed different simulation models to investigate the wheel climb phenomenon. Such an approach, however, lacks models that can be used to evaluate the wheel climb derailment phenomenon and other added complexities, such as the environmental factors associated with it. Thus, it is far more challenging to rely on the abundance of data available to make groundbreaking decisions and improve the safety norms of the industry. To pave a new path, O’Shea and Shabana [100] built a fully non-linear multibody system wheel climb derailment model to study the effects of large AoA. An additional analytical model is developed to compare and analyze the results. An additional analytical model is developed to compare and analyze the results. The MBS model was developed using SIGMA/SAMS software. An elastic contact problem is used to model the wheel–rail contact phenomenon with an AAR-1B wheel and a 140lb rail. The AREMA rail profile is used in the model, which depends heavily on geometric idealization. A constant lateral force is applied to initialize lateral displacement in the model. Hertzian contact theories are used to determine the dimensions of the contact ellipse. Kalker’s USETAB is used for creep force measurements [42].

The authors provide an in-depth explanation of the nuances of the wheel climb process. This analytical model used partial results from the MBS model. Figure 21 shows the L/V results from the MBS model, which are compared to Nadal's derailment criteria. The authors say that Nadal's derailment criteria are not conservative, as their predicted value begins lower than Nadal's predictions. The L/V increases with time and goes above Nadal's limit of 0.95. Then, if the limit falls below the maximum, a semi-analytic model was formulated for comparative purposes. Conclusions are drawn that Nadal's criteria are not conservative, which contradicts the general assumptions considered. Although this requires further investigation and model verification by other researchers in the same field, the effort is commendable, as it provides another direction to the current understanding of the L/V force criteria.

In this aspect, O'Shea and Shabana [101] also studied the three-point contact phenomenon (the two-point flange contact on the leading outer rail and the tread contact on the inner rail). The investigators formulate the L/V criteria and Nadal's derailment limit. They conclude that an AoA is required to identify the contact point geometrically. A clear distinction is made between the importance of the contact angle and the flange angle. With this, the authors question the accuracy of Nadal's criteria as it solely involves the flange contact angle. The authors emphasize the importance of geometry in their investigation of the complex wheel–rail interaction phenomenon, specifically wheel climb derailment. Relying mainly on models, the mentioned studies can be extended to potentially integrate real-time test validation using IWSs. Alternatively, they can employ dynamic simulations that better incorporate friction variation and track irregularities. They can also assess the scalability of the models to different vehicle types and operating conditions.

The combined efforts mark a significant step toward bridging experimental observations with numerical predictions of flange climb derailment. These studies explore how wheel geometry, contact angle, and coefficient of friction interact during the critical stages of wheel lift and re-contact. Laboratory-scale rigs and model bogies have provided valuable visualization of wheel climb onset and post-climb behavior, while simulation tools have helped interpret the underlying force dynamics. However, scaling constraints and simplified boundary conditions often limit the translation of these findings to full-scale, high-speed systems. Many experiments capture quasi-steady climb processes, whereas real derailments occur under transient and multi-axial load interactions. Similarly, several numerical models assume ideal contact geometry or uniform material behavior, which can overlook micro-scale effects like slip reversal or contact saturation. Despite these limitations, this combined experimental–numerical body of work has deepened the mechanistic understanding of flange climb and revealed

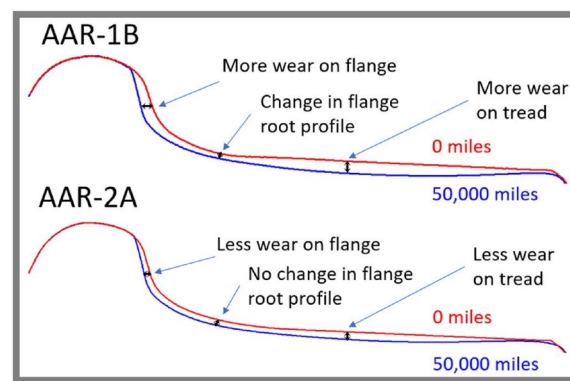


Fig. 22 AAR-1B versus AAR-2A wheel profiles

that derailment thresholds are highly sensitive to geometry, adhesion, and local contact transitions. Continued integration of scaled testing with validated dynamic simulations will be essential to capture the true progression of flange climb events and refine derailment prediction frameworks.

5.2 Wheel and rail design

Wheel and rail design play an important role in rolling contact mechanics. Based on a thorough review of contact forces and models, choosing the right wheel and rail profiles impacts the results. Broadly, wheel–rail design includes both profile and material choices. Wheels are complex; the tread has a low-level conicity; as you move toward the throat, the angle increases, reaching the highest—often between 65° and 75° —at the flange. Generally, rail profile guidelines follow a simple pattern, with a large radius at the head—mainly for wheel tread contact—and a smaller radius at the gauge corners where flange contact might occur. In 2005, the Transportation Research Board, working with the National Research Council, conducted a detailed survey to develop comprehensive maintenance guidelines for flange climb derailment prevention and wheel or rail profile management [102]. The survey included six transit systems. The findings highlight the importance of wheel and rail profiles in maintaining smoother system operation. Here are some key points related to profile standards, summarizing common concerns and necessary guidelines:

- The risk of derailment increases when there is a low flange angle and when the wheel profiles are rough or irregular due to several causes.
- For the design of the flange angle, at least 75° is recommended.
- Severe two-point contact produces poor steering capabilities, thus increasing the risk of derailments.

- Friction management between the wheel and the rail needs further exploration.
- A new wheel and rail profile requires careful programming for evaluating various running conditions.

Wheel and rail profiles are crucial in determining contact point locations and several other factors, including wear and wear rate, surface damage and deformation, maintenance costs, and curving ability. Iwnicki [103] examined how the profile influences damage on the wheel and rail, focusing on wear and rolling contact fatigue (RCF). The Manchester Metropolitan University (MMU) software [104] was used for simulations. The author emphasizes the difficulty of designing an optimized wheel profile that balances stability and derailment resistance while minimizing wear and RCF initiation. The proposed method considers these factors for profile optimization. The Association of American Railroads (AAR) introduced the AAR-2A wheel profile as part of its research initiatives, replacing the AAR-1B profile on January 1, 2019, and integrating it into revenue service [105]. The Transportation Technology Center Inc. (TTCI) conducted comprehensive analysis, modeling, and field tests. This new profile was developed using historical wear pattern data from the AAR-1B profile (Fig. 22), specifically to prevent hard two-point contact during curve negotiation, which results in high lateral forces. The conformal contact on the AAR-2A profile improved train stability during curves. Additional findings indicate nearly a 58% reduction

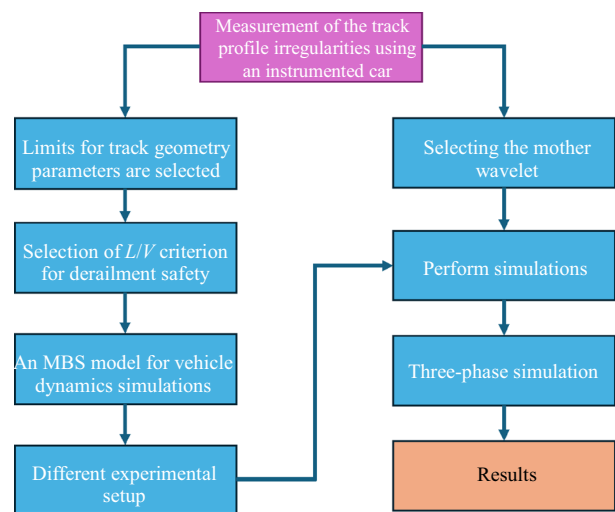


Fig. 24 Experimental study adopted by Costa et al. [112] to employ track irregularities and simulate derailment limits

in flange wear for track curves with curvatures from 1° to 6° [106]. Ongoing research continues to explore contact patch estimation through both experimental and modeling methods [39, 107].

Mohammadzadeh et al. [108] estimate the probability of train derailment by modifying rail profiles. Nadal's derailment coefficient is used in their study to measure derailment risk. Figure 23 (method A) shows the experimental approach

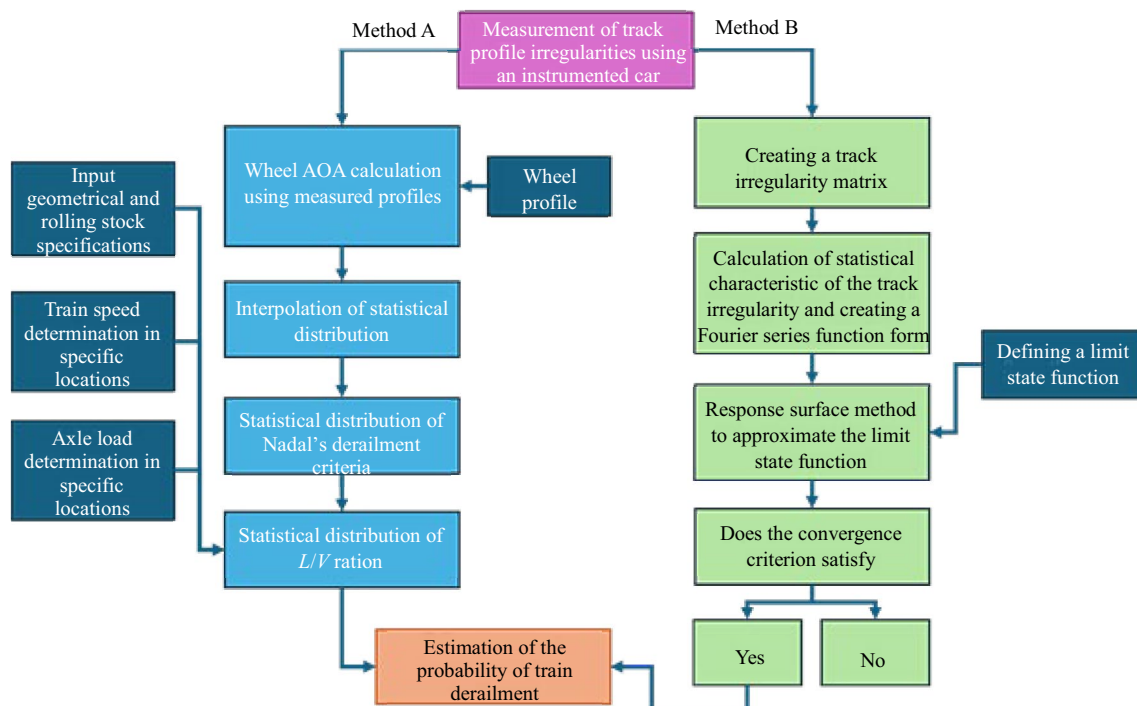


Fig. 23 Summary of the two methods proposed to estimate the probability of derailment (adopted from [108, 109])

considered by the author. Using this numerical method, the derailment risk can be calculated. For a set of three curved track cases examined by the authors, the model was able to predict the hierarchy of track maintenance needed to prevent derailment. In a similar attempt (method B in Fig. 23), the same authors, Mohammadzadeh and Ghahremani [109], estimate the derailment probability using random track geometric irregularities. SIMPACK was used as a tool to simulate the model. Here, Nadal's derailment coefficient proves useful, as the authors apply the limit state function to predict derailment. Successful validation of the simulations was carried out. This can assist other researchers in applying this model for further parametric evaluation and understanding the correlation between track irregularities and derailment risk.

Cinal et al. [110] employ a nonlinear dynamic 27-degree-of-freedom model with a select wheel and rail profile combination, incorporating additional random track irregularities. With the help of power spectral density (PSD), the spatial frequency of the track irregularities was used. The authors suggest that this model can make contributions to a better understanding of vertical and lateral track irregularities. The authors conclude that the first three peaks observed are characteristic peaks unaffected by the last stiffness, vertical damping in the primary suspension. Further evaluation is required to investigate in depth the use of such models using PSD to observe hunting phenomena. Costa et al. [111] employ wavelet analysis and vehicle dynamic simulations to filter track irregularities into different wavelengths and reconstruct these with other wavelets. An illustration of the experimental approach considered is highlighted in Fig. 24. They compare how this affects the L/V ratio. The design comprises simulations from which various patterns of L/V arise. The authors suggest how the intrinsic shape of the wavelet plays an important role. The similarity of the wavelet shape and the defects can be used for further targeted maintenance actions. Some assumptions have been used by the authors, which must be investigated; for example, in the model, all the tests performed have no chance of derailment based on the limits considered.

Gullers et al. [112] investigated the impact of track irregularities and the estimation of wheel–rail contact forces using in-field measurements. Four different rail irregularity classes were identified, requiring high-frequency dynamics to evaluate contact forces.

There have been limited studies on the importance of wheel and rail profile designs for derailment analysis or mitigation. While there is abundant research on the tribological aspects of the wheel–rail interface, little progress has been made in connecting it to derailments. Thus, future research must heavily employ analytical and numerical models to

replicate several wheel and rail profiles to iteratively study their impact on derailment criteria and contact forces.

6 Key findings and knowledge gap identified

The preceding review highlights the vast research on derailment mechanisms induced by wheel–rail interaction. The focus has been on experimental and simulation-based studies. This study presents a detailed analysis of several consistent methodologies and conclusions, highlighting specific knowledge gaps that remain.

Several researchers have used different methods to understand the complex contact mechanics at the wheel–rail interface. Some have employed instrumented wheelsets, instrumented rails, wayside applicators or devices, and roller rigs [84, 85 113]. These studies emphasize the impact of contact forces and, consequently, the derailment criteria. Using these methods, many authors can measure contact forces and either validate Nadal's derailment limit or demonstrate that it is too conservative. Although conservative, this limit appears to be central to many research developments. Investigators have also used advanced modeling tools to simulate vehicle system dynamics with greater accuracy. Some field measurements are also utilized to help validate models that simulate specific environmental conditions. The progress in wheel–rail contact models also shows high reliability in confirming the contact forces, contact shape, and contact points predicted by these models. A comprehensive review of such models is available here [45].

At the end of the review, the significance of wheel and rail profiles, their design, and overall maintenance practices that can help minimize the risk of derailment are emphasized. The wheel and rail profiles significantly impact wear and surface damage resulting from contact stresses, high temperatures, and other factors. The introduction of the new AAR-2A wheel profile by the American Revenue Service highlights the impact of innovative designs on achieving good curving dynamics. Several researchers have investigated ways to optimize wheel and rail profile design based on observed or measured track irregularities by developing optimization models using various tools that enable faster and more accurate calculations. It can be said that a good balance of experimental and theoretical research is currently underway. However, there remains a need for more hybrid studies that can be scaled for practical application in the field.

Despite significant progress in understanding wheel–rail contact mechanics and derailment phenomena, several gaps still limit a comprehensive understanding of derailment mechanisms. Beyond the extensively studied wheel climb derailments, derailment coefficients, and contact force measurements, other forms of derailments

caused by the wheel–rail interaction, such as wheel sliding and wheel impact, have received comparatively little attention, particularly from experimental and simulation standpoints. Wheel sliding derailments typically arise from a low coefficient of friction, often due to third-body contaminations, such as flange grease migration to the top of the rail or leaf contamination, both significantly reducing adhesion. Conversely, wheel impact derailments result from sudden dynamic contact forces caused by wheel flats, out-of-round wheels, damaged rail surfaces, or irregular track geometry. Although some studies have evaluated these effects, individually, very few have directly correlated them with derailment mechanisms or quantified their contribution to derailment risk. Numerous research initiatives are underway to investigate the wheel and rail defects, wear, lateral impact, traction enhancers, gauge widening, track buckling, and rail head contamination [114–124]. Few studies address the causes of wear [125–127] and RCF [129–130], but there is no unified understanding of how such defects contribute to derailment risk. This lack of integrated research highlights a crucial opportunity for future investigations to bridge laboratory, field, and numerical theories.

Moreover, while Nadal's derailment coefficient remains the industry's most widely applied safety limit, its conservative nature stems from the assumption that derailment primarily occurs at the flange angle limit. In reality, the contact angle, which varies with geometry, wear, and load transfer, may provide a more physically representative indicator of derailment onset. However, accurately estimating this angle under dynamic operating conditions remains challenging, mainly when multi-point or conformal contact occurs. To better understand and potentially reduce such effects, several researchers have evaluated wheel and rail profiles that can reduce WUL and reduce the effect of multi-point or conformal contact.

Collectively, these insights underline the need for comprehensive hybrid research frameworks that integrate high-fidelity simulations, controlled laboratory rigs, and field-scale validation. It is recommended that future studies incorporate contact geometry estimation, force measurement, and uncertainty quantification to develop adaptable derailment models that capture real-world variability. Leveraging modern tools such as machine learning (ML), data-driven modeling, and sensor-based continuous monitoring could further strengthen predictive capabilities and accelerate the implementation of derailment prevention strategies.

The authors believe that the increasing relevance of artificial intelligence (AI) and ML will speed up the development of technologies to advance these studies. The railroad industry, long rooted in empirical practices and century-old engineering wisdom, now stands at the brink of a data-driven transformation. With abundant data from simulations, IWSs, and continuous

track-monitoring systems, AI enables converting this wealth of data into actionable insights. By learning from both field and laboratory datasets, intelligent algorithms can refine and tune long-standing simulation models, reducing uncertainties and bridging the gap between theory and practice. When coupled with physics-based frameworks, these models can form digital twins capable of predicting derailment precursors, identifying evolving contact geometries, or flagging track irregularities before it's too late. We're taking the best of both worlds: using modern data and analytics to improve upon our proven mechanical understanding.

This literature review, therefore, serves not only as a critical synthesis of past achievements but also as a launching point for global researchers to rethink derailment studies in a more interconnected and multidisciplinary way. The authors hope that addressing these overlooked aspects and fostering data-driven experimentation will improve rail safety and operational efficiency.

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